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Topical Review

Numerical modelling of wire arc additive manufacturing: methods, status, trends, and opportunities

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Abstract

Wire arc additive manufacturing (WAAM) is a metal additive manufacturing (AM) process attracting interest from the automotive, defence, aerospace, architecture/engineering/construction and other industries because of its ability to manufacture large metal parts cost-effectively. Nevertheless, problems such as part defects and process efficiency remain, and consequently, efforts to improve WAAM are continuing. The WAAM process involves physical phenomena that include fluid flow, heat transfer, phase changes including melting, solidification and vaporization, multi-phase interactions, and deformations resulting from residual stresses—so obtaining a complete understanding is challenging. While numerical modelling is widely used to understand and assist with developing processes, modelling of AM processes such as WAAM is sophisticated because of their multi-physics and multiscale nature. This review addresses the existing and likely future roles of numerical modelling in advancing WAAM technology. Consideration is given to the known problems with WAAM, the different types of numerical modelling, including computational fluid dynamics, the finite element method, and smoothed particle hydrodynamics, and their potential to address persistent issues. Additionally, this review seeks to provide an understanding of the physics associated with the WAAM process, examines the trends in the development of WAAM technology, and recommends possible future directions. These include the combination of different physics-based modelling approaches to overcome their individual shortcomings, and the inclusion of modelling as part of a digital twin of the WAAM process.

Keywords: wire arc additive manufacturing, computational fluid dynamics, smoothed particles hydrodynamics, finite element analysis, digital twin, arc welding, arc physics

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Nomenclature

$\boldsymbol{A}$	Magnetic potential
A_x	Cross-sectional area along the x direction
a_s	Activity of the solute
B	Magnetic field strength
c_p	Specific heat (at constant pressure)
E	Elastic modulus
e	Charge on an electron
F	External force vector
F_arc	Force of the arc
F_b	Buoyancy force
$F_{\mathrm{em},z}$	Approximate axial electromagnetic force
F_x, F_y, F_z	Lorentz forces in x, y and z directions respectively
F_γ	Surface tension force
f_f	Front distribution fraction
f_l	Liquid fraction
f_r	Rear distribution fraction
$\boldsymbol{g}$	Acceleration due to gravity
H_vap	Molar heat of vaporization
H^o	Standard heat of adsorption
h	Interpolation length
h_a	Specific enthalpy of particle a
h_s	Specific enthalpy
I	Arc current
J_vap	Metal vapour flux
$\boldsymbol{j}$	Current density
j_e	Electron current density
KE_droplet	Impact kinetic energy of the droplet
k	Thermal conductivity
k_a, k_b	Thermal conductivities of particles a and b
k_B	Boltzmann's constant
k_s	Stiffness
L	Latent heat of fusion
L_0	Substrate thickness
Δl	Mesh dimension perpendicular to surface
m_a, m_b	Masses of particles a and b
m_M	Atomic mass of metal
$\hat{\boldsymbol{n}}$	Surface unit normal vector
n_a	Unit normal to particle a
P	Pressure
P_arc	Pressure of the arc
P_atm	Atmospheric pressure
P_vap	Metal vapour pressure
P_∞	Pressure at a small distance from electrode
q_x	Thermal load in x direction
R_cathode, R_anode	Radius of the arc at the cathode and anode
R_droplet	Radius of molten metal droplet
R_g	Ideal gas constant
r	Radius, given by $r = \sqrt{(x - x_0 - vt)^2 + (y - y_0)^2}$

r_{ab}	Relative position of particle a and b, $r_a - r_b$
$\boldsymbol{r}_b$	Position of particle b
r_{eff}	Effective radius
S_{bb}	Heat flux transferred by blackbody radiation
S_{droplet}	Heat flux transferred by droplets
S_{ht}	Heat transfer source term
$S_{ht\,\zeta}$	Double ellipsoidal arc heat source
S_{m}	Momentum source term
S_{vap}	Heat flux transferred by vaporization
T	Reference temperature
T_{ab}	Relative temperature of particles a and b, $T_a - T_b$
T_{b}	Boiling temperature of the metal
T_{L}	Local temperature
T_{l}	Liquid metal temperature
T_{liq}	Liquidus temperature
T_{m}	Melting temperature
T_{sol}	Solidus temperature
U	Net radiative emission coefficient
$\boldsymbol{u}$	Nodal displacement vector
V	Arc voltage
V_{s}	Sheath voltage
$\boldsymbol{V}_{\text{w}}$	Travel velocity of the wire, or welding velocity
$\boldsymbol{v}$	Flow velocity
$\boldsymbol{v}_a$, $\boldsymbol{v}_b$	Velocities of particles a and b
v_{droplet}	Speed of molten metal droplet
v_{p}	Arc plasma velocity
x	Distance from the origin along x-axis
x_a, y_a, z_a	Cartesian distances to the arc centre along x-, y- and z-axes
x_0	Origin of the x-axis
y	Distance from the origin along y-axis
y_0	Origin of the y-axis
z	Distance from the origin along z-axis
z_0	Origin of the z-axis
z_{perp}	Coordinate perpendicular to the melt pool surface
β	Thermal expansion coefficient
Γ_s	Adsorption at saturation point
γ	Surface tension coefficient
γ_{bulk}	Bulk modulus
$\gamma_{\text{m}}^{\text{o}}$	Surface tension coefficient at the melting temperature
δ	Prevention factor for zero denominator
ε_x, ε_y	Strain at FEA mesh node in x and y directions
κ	Surface curvature
μ_{m}	Substrate permeability
μ_{p}	Dynamic viscosity of the arc plasma
μ_0	Permeability of free space
ν	Poisson ratio
$\boldsymbol{\Pi}_{ab}$	Viscosity used to generate a shear
ρ	Mass density
ρ_a, ρ_b	Densities of particles a and b
σ	Electrical conductivity
$\boldsymbol{\sigma}_a$, $\boldsymbol{\sigma}_b$	Stress tensors of particles a and b
σ_j	Effective radius of the current density distribution
σ_{p}	Radius over which the arc pressure acts
σ_x, σ_ψ	Normal stress profile in x and ψ directions
$\boldsymbol{\tau}$	Stress tensor
τ_a	Arc shear stress
τ_{xy}	Shear stress
Φ_a	Rate of cooling due to convection or radiation of particle at free surface
φ	Electric potential
$\varphi_{\text{cathode}}, \varphi_{\text{anode}}$	Work function of cathode and anode materials
Ψ	Entropy factor
$[C]$	Heat capacity matrix
$[K]$	Conductivity matrix
$\{q\}$	Thermal load matrix
$\{T\}$	Nodal temperature matrix
$\{\dot{T}\}$	Time rate of change of temperature at the FEA mesh node

1. Introduction

1.1. Wire–arc additive manufacturing (WAAM)

Metal additive manufacturing (AM) uses an energy source such as a laser, electron beam or arc plasma to melt and deposit a metal powder or wire layer by layer. Metal AM processes fall under the AM classifications of powder-bed fusion, directed energy deposition and binder jetting [1]. They are attractive to manufacturers in sectors including aerospace, biomedical, automotive, defence and energy because of their ability to produce bespoke parts with low waste, often with complex designs that cannot be manufactured by traditional methods.

Direct energy deposition processes, which include blown powder, laser-wire, electron-beam-wire and WAAM, use focused thermal energy to fuse materials by melting as they are deposited [1]. They made up approximately 16% of the market share in the AM industry in 2020 [2, 3].

WAAM, also known as wire–arc direct energy deposition, has evolved from arc welding and uses an electric arc as the heat source to melt a metal wire. Like other metal AM processes, WAAM provides substantial freedom in part shape and scale while reducing material waste. As well as building parts layer by layer, WAAM can be used directly to repair metal parts. WAAM is generally carried out with a robot arm integrated with an automated system linked to computer-aided drawing (CAD) software, which is suited to design customization. WAAM has a much higher production rate than powder-based AM processes and thus is used to produce larger parts, especially aluminium-alloy-based components. As a result, the aerospace, shipbuilding and automobile industries are leading beneficiaries of WAAM. Despite its benefits, several problems affect the process efficiency and quality of the fabricated product. Also, the resolution and complexity of the parts produced by WAAM are typically lower than those obtained using powder-based AM processes.

Several review papers have covered aspects of WAAM, including the design and implementation of WAAM systems, *in-situ* process monitoring and metrology, process control and sensing [4–6]. Wu *et al* reviewed the properties of the metal parts fabricated by WAAM, including common defects and their prevalence for different materials, and methods for quality improvement and defect reduction [7]. Norrish *et al* recently provided an excellent overview of the physics underlying the WAAM process, and reviewed numerical modelling, process control and materials considerations [8].

Barath Kumar *et al* considered finite element analysis (FEA) simulations as part of a review focused on the properties of the metal parts produced by WAAM [9]. Several other reviews have considered computational modelling of metal AM but have concentrated on powder-based processes [10–16]. Srivastava *et al* review was devoted to the computational modelling of WAAM but was heavily weighted towards FEA for calculating residual stress in metal parts [17]. Sampaio *et al* recently reviewed modelling of WAAM, again focusing on macro-scale FEA; they also briefly considered machine learning (ML) approaches [18]. Accordingly, this review aims to present a more detailed and comprehensive perspective of numerical modelling of WAAM, including fluid models that consider the arc plasma and melt pool, and to provide proposals for the future development of WAAM modelling.

1.2. Numerical modelling

The applications of numerical modelling continue to broaden as computing capacity becomes more powerful and new and improved numerical techniques and software are developed. For example, graphics processing units allow massive parallelization, the advent of cloud computing has increased the availability of computing resources, commercial software packages offer new and improved capabilities, and new numerical approaches continue to be developed [19].

Numerical modelling and simulation methods have been applied extensively to understanding and developing metal AM, including WAAM, in the past decade [20–23]. The finite element method (FEM) is widely used to track heat transfer during metal AM and is an essential approach for the structural analysis of AM products [24]. Computational fluid dynamics (CFDs), the lattice Boltzmann method and smoothed particle hydrodynamics (SPHs) have all been applied to modelling the melt pool [10, 23, 25, 26]. The discrete element method (DEM) is used to predict powder spreading in powder-based AM methods [10, 23, 27].

In common with all metal AM processes, WAAM involves many physical phenomena, including fluid flow, heat transfer, phase changes such as melting, solidification and vaporization, and multi-phase interactions. Modelling of WAAM is simplified by the fact that a wire rather than powder is used as the metal source. However, the use of an electric arc rather than a laser or electron beam to melt the metal is a significant complication. Moreover, the large (up to metres) scale of parts produced by WAAM, compared to the small (mm) scale of the wire, arc and melt pool and finer-again scale of the metal microstructure, presents major challenges. Treating all the relevant phenomena would require a highly complex multi-scale and multi-physics model, and current efforts have been restricted to only some of the physical phenomena and scales.

A further demand on computational models arises from the increasing investment in developing digital twins of manufacturing processes such as WAAM. Computational models will form a critical component of many such digital twins. However, coupling the digital twin to the physical process requires real-time simulation, which is not attainable with standard approaches such as CFD, and since a highly accurate prediction requiring the vast amount of available data is desirable, ML based or other surrogate models are likely to be required.

1.3. Contents of the review

We begin this review by explaining the basic concepts of WAAM, including the types of WAAM and the problems with the process, in section 2. We then outline the rationale for the computational modelling of WAAM in section 3, and outline the different computational methods that can be adopted to model WAAM, including their advantages and limitations, in section 4. We review the literature on numerical modelling on WAAM to evaluate how various simulation approaches can tackle the problems of WAAM and present demonstrations of each simulation method in section 5. In section 6, we analyse current trends in the computational modelling of WAAM and the opportunities these present, followed by our proposals for the future development of WAAM modelling, including combined approaches that could be advantageous in capturing the multi-scale and multi-physics requirements and an analysis of how modelling and experiment could be combined to develop digital twins of WAAM.

2. Basic concepts

2.1. What is WAAM?

WAAM employs an electric arc as a heat source to melt a metal wire, depositing molten metal layer by layer on a substrate to create a metal part. WAAM evolved from industrial shape welding, first reported in the early 1980s, where the original production of complete industrial shape welding parts was large nuclear pressure vessel parts developed in Germany [8]. A WAAM system, as shown in figure 1, consists of a welding torch attached to a robot arm, power and shielding gas supplies to the welding torch, a feed wire that is melted by the electric arc, a substrate on which molten metal is deposited to build the part, and a control system. Typically, a CAD file containing a three-dimensional (3D) representation of the desired part will be produced and input to software that slices the 3D model into the layers to be deposited. Path programming is then conducted to plan and optimize the welding path [28]. Deposition parameters, including deposition rate, path speed, mass flow rate and the arc current, are defined and input to the system. A computer numerical control code is generated based on the deposition parameters, and the system begins to additively manufacture layer-by-layer, with the molten metal droplet detached from the tip of the feed wire deposited on the substrate. The mode of transfer of the molten metal depends on the welding and deposition process conditions [29] and can significantly influence the quality of the product. When the deposition is complete, post-processing, such as milling, cutting and polishing, and post-heat treatment, can be conducted to further optimize the part properties [30].

WAAM is generally more cost-effective than other metal AM processes. The power density provided by the arc is lower

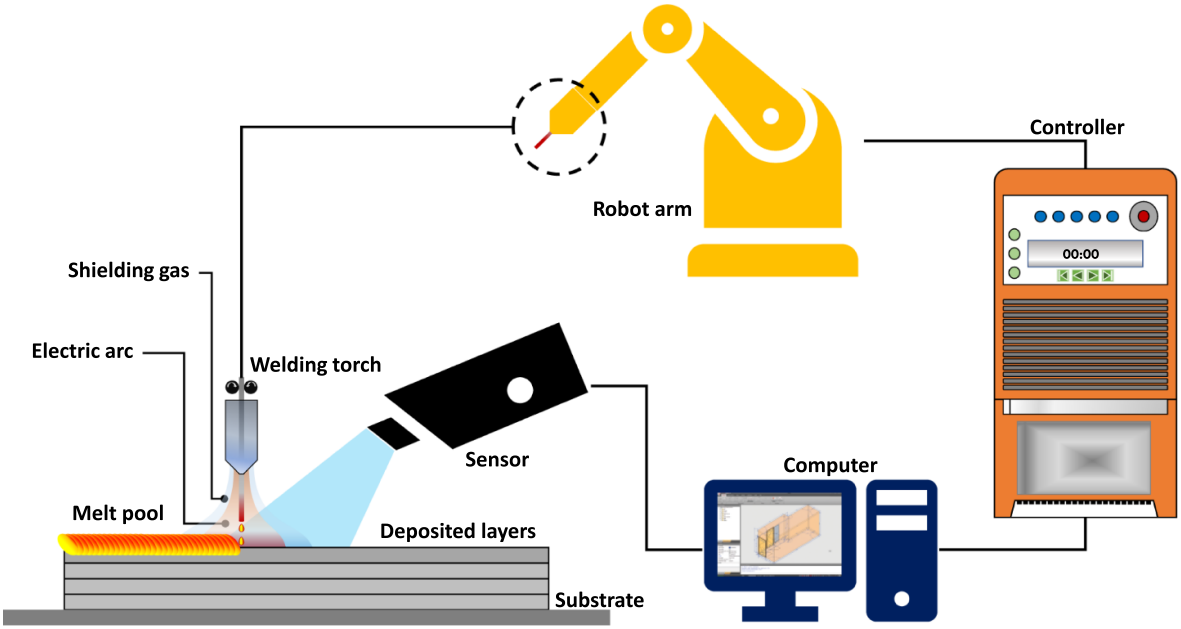


Figure 1. Schematic of a WAAM system.

than that offered by the laser or electron beam in powder-bed fusion or other types of direct energy deposition, resulting in comparatively lower surface finish quality. However, the arc power is higher, allowing a greater material deposition rate, faster processing speed, and comparatively low cost [31]. The arc generation efficiency is also much higher; the energy required to produce the arc in WAAM is reported to be approximately one-tenth and one-thirtieth, respectively, of those for laser and electron-beam generation. Moreover, the cost of the feed wire is between one-third and one-half of metal powder in the same material [32]. Due to these advantages, WAAM is being adopted to manufacture large parts in several industries, including aerospace, shipbuilding, energy and defence, particularly to manufacture high-density parts from costly materials when surface-shape resolution is not critical. WAAM is also used for the repair of damaged or aged parts [33]. This feature is well aligned with the circular economy concept and is an example of a sustainable manufacturing process.

2.2. *Types of WAAM*

Figure 2 shows schematics of WAAM types. WAAM is developed from arc welding techniques and can be categorized based on the nature of the arc source that is employed to melt the feed material [7]. There are three main types of WAAM: (a) gas metal arc welding (GMAW) [5], (b) gas tungsten arc welding (GTAW) [34] and (c) plasma arc welding (PAW) [35].

2.2.1. GMAW. GMAW is also known as metal inert gas or metal active gas welding, depending on whether the shielding gas is inert. As shown in figure 2(a), an electric arc is generated at the tip of a feed wire in GMAW, so the feed wire serves as an electrode (usually the anode). GMAW provides a high deposition rate, but since the melting rate of the wire is determined by the arc current, the wire feed rate and the arc current are coupled. The current also determines the metal transfer mode (e.g. short-circuit, globular or spray transfer). The arc can be unstable, resulting in inferior surface quality and a high volume of weld fume and spatter. Cold metal transfer (CMT), a type of short-circuit transfer in which the feed wire is retracted during the short-circuiting phase [36], provides improved arc stability, significantly reducing spatter. The relatively low thermal input during metal transfer reduces heat-induced distortion.

Several variants of GMAW exist. Tandem GMAW, which uses two feed wires and arcs, can improve the travel speed and deposition rate [37]. Double-electrode GMAW, which adds a GTAW torch to the GMAW system, with some of the current flowing from the GMAW electrode to the GTAW electrode, reduces the heat flux to the substrate [38].

2.2.2. GTAW. GTAW, also known as tungsten inert gas welding, uses a non-consumable tungsten electrode (usually the cathode) to generate an arc, and the feed wire is introduced to the melt pool. Unlike GMAW, the feed wire in GTAW is separate from the electrode, being fed at a small angle and adjacent to the substrate, as shown in figure 2(b). This has the advantage of allowing the wire feed rate and the arc current to be separately controlled. Twin-wire feeding also facilitates the additive manufacture of intermetallic materials with a compositional gradation [37].

2.2.3. PAW. Figure 2(c) shows a schematic of PAW. This is similar to GTAW in that the electric arc is generated between a non-consumable tungsten electrode (usually the cathode) and the metal substrate. However, an outer nozzle provides an

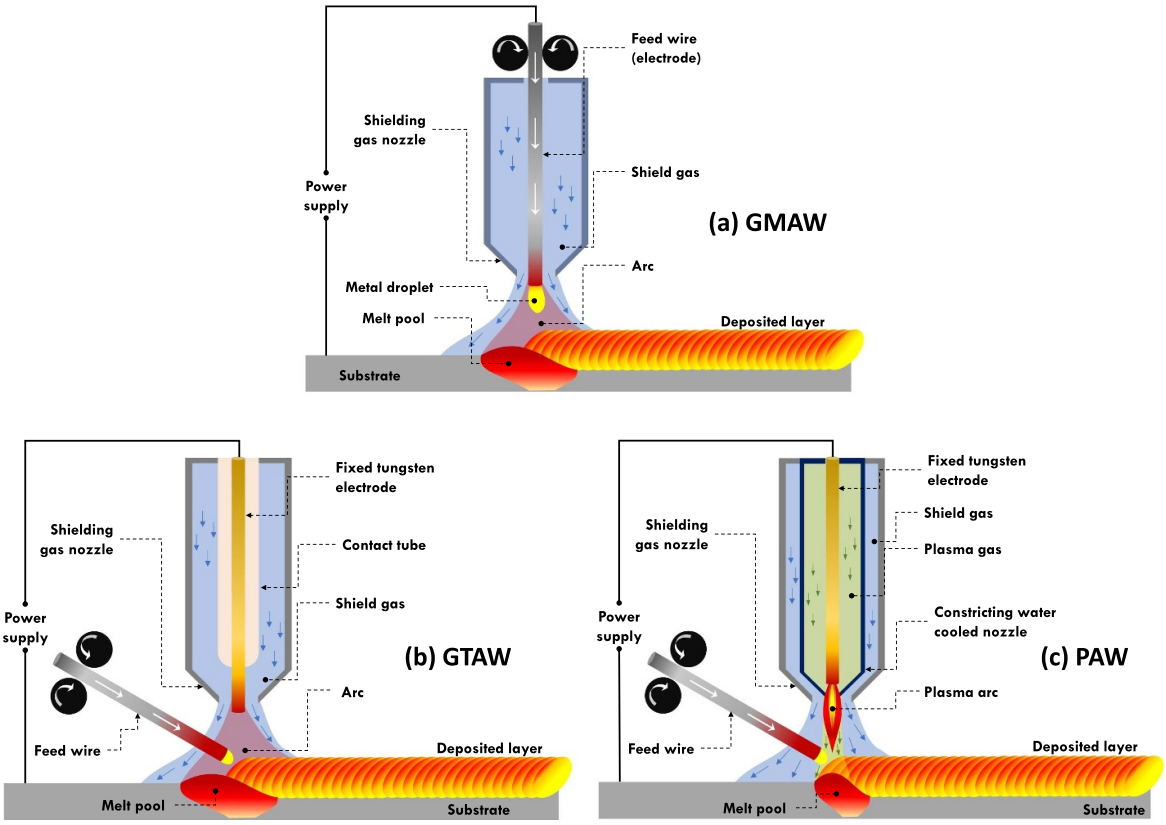


Figure 2. Schematics of WAAM types: (a) GMAW, (b) GTAW and (c) PAW.

additional shielding gas stream around the arc region, which leads to a more constricted arc [39] that transfers a more focused heat flux to the workpiece [40]. This may reduce the size of the heat-affected zone and facilitate smaller-size welds, making microscale WAAM, namely micro-PAW, possible.

2.2.4. Deposition rates. The deposition rates of GMAW, GTAW and PAW are approximately 3–4 kg h^{-1}, 1–2 kg h^{-1} and 2–4 kg h^{-1}, respectively. Tandem GMAW allows the rate to be increased to 8 kg h^{-1} since it adopts two consumable wire electrodes, while the deposition rate for CMT is a little lower than standard GMAW, about 2–3 kg h^{-1}.

2.3. WAAM problems

We consider the problems associated with WAAM in some detail since one of the main purposes of computational modelling is to understand and reduce these problems. Overcoming or minimizing their effects is essential for quality assurance and process reliability. Most of the problems are inherited from arc welding, but we also consider those specific to WAAM.

2.3.1. Heat-induced problems. WAAM uses a substantial amount of localized heat energy for material fusion, leading to defects such as undesirable microstructures, residual stresses, and deformation [41]. Another problem induced by heat is metal vapour, which is a precursor for welding fume and can affect the environmental sustainability of WAAM [42].

The heat transferred by the arc and the molten metal leads to a highly non-uniform temperature distribution in the substrate, with iterative heating and cooling cycles occurring as successive layers are deposited. The resulting non-uniform thermal expansion and contraction of the metal leads to the development of residual stresses in the parts produced by WAAM, which in turn causes distortion [5, 43]. Residual stress and distortion can lead to parts being out of design tolerance and having poor mechanical properties [37]. The amount of residual stress and distortion depends on many factors, including the heat flux from the arc and transferred metal, the thermophysical and thermomechanical properties of the metal, the part geometry, and the deposition path, since the pattern traversed can have a large impact on the peak temperature and heating and cooling rates. Furthermore, the number of deposited layers affects the heat accumulation, with an increasing number of deposited layers leading to reduced heat conduction toward the substrate, which becomes more uniform over time [44, 45]. The reduced heat loss slows the cooling of the melt-pool region.

The thermal history likewise determines the microstructure of parts produced by WAAM [46], which again depends on the heat flux to the substrate, the thermophysical properties of the metal, and the part's geometry. The microstructure also depends on the composition of the metal alloy, with the repeated heating and cooling of the deposited metal likely

forming non-equilibrium phases [47]. The reduced heat conduction towards the substrate as layers are added can lead to a non-uniform microstructure, which is more likely to be columnar at the bottom of the part and equiaxed at the top, resulting in inconsistent mechanical properties at different locations [48, 49]. Aluminium and magnesium alloys are particularly prone to the formation of a non-uniform microstructure with coarse columnar grains [50].

Post-processing and additional processing methods can be used to partially relieve residual stresses and refine the microstructure. Rolling is an example of a process that can be used during production [51]. A roller can be applied to the top of the build between passes or to the side of the build. Recent investigations of the effectiveness of the rolling process in improving microstructure and residual stress [52, 53] have demonstrated a significant reduction in the peak residual stress and distortion [52]. However, all post-processing adds complexity and cost, so reducing the residual stress and optimizing the microstructure produced during the WAAM process remain important goals. Other approaches include active cooling during manufacture, including immersion of the part in a water bath and air or aerosol jet cooling. These approaches are faster than pausing between layers [54]. Immersion in a water bath and air-jet and aerosol-jet cooling have all been used in experimental trials. Forced air cooling, and possibly the other methods, are used in manufacturing. In the Eulerian framework, a water bath can be established as a fixed-temperature boundary condition. If water heating is to be considered, then a heat-flux boundary condition can be used, or the water bath can be included in the computational domain. Cooling via an air or aerosol jet can also be modelled in an Eulerian framework by using a heat flux boundary condition, with the heat flux coefficient increased to account for the forced-air or aerosol cooling. All these processes are amenable to computational modelling.

Metal vapour is generated by the evaporation of molten metal from the melt pool, droplets and the tip of the feed wire [42]. The presence of metal vapour has a major influence on the arc plasma, predominantly by increasing radiative emission and decreasing the temperature at which the arc conducts electricity. The former effect cools the arc, while the latter increases the area of arc attachment to the substrate, both reducing the heat flux transferred to the substrate and the overall performance efficiency. Murphy indicated that iron has the strongest radiative emission, followed by copper, while aluminium and silicon have comparatively weak radiative emission [42].

In addition, metal vapour is the primary source of welding fume, formed when the metal vapour is transported to the fringes of the arc plasma, leading to nucleation and growth of metal and metal oxide nanoparticles, which generally agglomerate to form chains [55]. GMAW-type WAAM produces comparatively more fume than other WAAM types [37]. Inhalation of fume is a recognized health risk for welders, with consequences including bronchial disease, cardiovascular disease and cancer [56]. While the production of metal vapour is inevitable, efforts to minimize its production, for example, by choosing the type of WAAM and the metal transfer process, should be considered.

2.3.2. Inertia-induced problems. Along with heat-induced problems, other problems occurring due to kinetic inertia can detrimentally affect the geometrical precision and surface quality of the part. These include the production of spatter (small metal particles produced from the melt pool or the molten tip of the wire), humping of the melt pool surface, lack of fusion, and deviation from the desired geometry.

Spatter is a common phenomenon, especially for GMAW-based WAAM. Spatter can occur due to the impact of metal droplets or the wire tip on the melt-pool region, often when fluctuations of the arc power lead to the arc temperature becoming too low to keep both the wire and melt pool molten. Asymmetries in the forces acting on metal transfer, including plasma drag, gravitational and electromagnetic forces, due to arc instabilities affected by the welding current and voltage can also lead to spatter [57, 58]. The effect of various shielding gases on spatter minimization has been studied experimentally [59–62]. Jafari *et al* highlighted that the CMT metal transfer mode, along with the addition of oxygen to the shielding gas mixture, notably decreased spatter production in comparison to conventional GMAW and the use of pure argon shielding gas, respectively [37]. Modelling of spatter is challenging, but studying the flow field of the shielding gas, the temperature of the wire tip and melt pool, and the metal transfer process can all provide valuable information.

Humping of the melt pool surface leads to a periodic undulation of the build surface. It is associated with strong backward flow in the melt pool [63] and has been attributed to premature solidification of the metal flowing from the front to the back of the melt pool induced by a capillary instability [64]. Humping can occur for high deposition rates in GMAW-type WAAM, typically for wire feed speeds above 2 m min^{-1} [65]. The high wire feed speeds correspond to large arc currents and faster travel speeds, but such conditions often lead to irregular melt-pool shapes, which in turn cause non-uniform bead width as well as humping defects [66–69]. Hybridization of the heat source in GMAW, such as the simultaneous application of a laser beam and the electric arc to the melt pool, has been shown to stabilize the arc, providing higher efficiency together with better wire feeding performance [70], with the wire feed speed before humping occurring increased to 3.5 m min^{-1} [65]. Such hybrid approaches add significant complexity and cost, so understanding humping through modelling is a worthwhile goal.

Lack of fusion, in which a deposited layer does not fully fuse with the substrate or previous layer, is an inertia-induced defect that can occur due to inadequate heat transfer to the build [71]. Lack of fusion can occur, for example, if the travel speed is too high or the arc does not attach stably to the build. Inadequate heat input causes poor penetration of the melt pool into the substrate or previously deposited layer.

Geometrical deviations often occur during the WAAM process, particularly for complex shapes [72]. There are four types of geometrical deviations: dimensional, shape, surface quality and positional deviations. Geometrical deviations of WAAM parts are often caused by distortion that has occurred due to the residual stress, in which case they are related to heat-induced defects. Geometrical departures also can occur when the

substrate is too weak to provide sufficient support to a heavy upper body.

2.3.3. Porosity. Porosity is a significant defect in WAAM [73]. Fu *et al* [74] identified poor deposition rate and porosity as the two primary technical challenges in the WAAM of aluminium alloys. Porosity affects the mechanical properties of components, shortening the lifecycle and, in severe cases, rendering the part unusable. There are many possible sources of porosity. Hydrogen porosity is a problem in aluminium alloys because hydrogen is much less soluble in solid than liquid aluminium [37, 75]. Hauser *et al* [76] studied the relationship of porosity to the shielding gas flow and observed that pores could be formed by entrainment of the ambient air into the shielding gas. High shielding gas flow rates may limit the escape of the entrained air from the melt pool by hastening its solidification. They also noted that the high dynamic pressure induced by the shielding gas surrounding the melt-pool surface could significantly contribute to porosity in the deposited metals through turbulent mixing [76]. Porosity is also induced by the presence of grease and/or moisture on the surface of filler wire, which produce pores in the deposited layers when these contaminants are incorporated into the melt pool [77, 78]. Biswal *et al* [79] reported that micro-porosity induced by contamination of the feedstock in titanium-alloy-based WAAM components significantly deteriorated the fatigue and fracture toughness, with the critical diameter of the pores observed to be approximately 100 μm. While macro-porosity can also be introduced by keyhole formation in PAW and laser welding, this is generally avoided in WAAM by choosing process parameters that avoid deep penetration and is, therefore, not a contributor to porosity [37].

Other defects, such as humping and inadequate fusion, are common challenges that can arise with any material. Consequently, particular materials cannot be associated with these defects.

Approaches such as pre-heating the wire (hot-WAAM or HWAAM) [74] and pulsing the arc at ultrasonic frequency [80, 81] have been shown to reduce porosity generation. However, such schemes add cost and complexity. Computational simulation of keyhole or porosity formation can help find conditions and parameters that minimize porosity in standard WAAM processes.

Problems arising during the WAAM process can stem from many factors, which can be difficult to identify experimentally. These factors can span multiple scales and can be driven by different physical processes. Numerical modelling can assist with identifying and understanding these processes and their effects, as discussed in the next section.

3. Importance of numerical modelling of WAAM

Numerical modelling is making increasingly valuable contributions to the development, optimization and operation of manufacturing processes. Modelling can be used to predict and understand complex multi-physics phenomena, including those that cause defects, and to assist the design, scale-up and scale-down of manufacturing processes and manufactured products. Moreover, process modelling is increasingly seen as contributing to part qualification and process certification and as a critical component of process digital twins.

3.1. Process improvement

Modelling can be applied to understand the link between process parameters and the defects considered in section 2.3. The use of simulations to optimize operational conditions and to determine how defects occur through two-way and real-time interactions is a core component of Industry 4.0 [19].

Many defects, including porosity, residual stress, and sub-optimal microstructure, are invisible. Others, such as humping, lack of fusion and part distortion, are apparent in the finished part but may require rectification. A computational model can allow the selection of process parameters that avoid or minimize defects without repeated trial-and-error experiments and testing. For example, CFD modelling has demonstrated that surface humping can occur due to a combination of the surface tension and the gravitational force acting on the substrate, causing a high backward flow momentum in the melt pool [37]. The development of residual stress and distortion, which results from the iterative heating and cooling of the substrate due to heating by the arc and droplets [82], can be simulated using FEA or SPH.

Validated models provide ‘virtual probes’ from which histories of field variables, such as temperature, may be obtained to replicate the process [83]. These probes—which complement actual, physical probes—can be as numerous as the computational model will allow (for example, the number of nodes in an FEA model) and can provide a complete picture, including information at locations as well as types where it is infeasible to place physical probes for measurement. They also allow the process model to be interrogated at a finer resolution to obtain further insights into the relationships between process parameters and results. Process models also allow the calculation of theoretical or hard-to-measure engineering quantities, such as stresses and strains, which are useful in guiding the optimization of processes [84].

Wu [85] proposed a framework for WAAM quality performance with three overarching aspects: process selection, feedstock optimization and post-process treatment. An accurate predictive model is an ideal tool for quality performance measurement and can ‘pre-print’ the component, saving time and resources. Process parameters such as the WAAM process type, wire design, path planning, shielding gas mix and heat source parameter coordination can be optimized. By doing so, the part geometry can be improved by reducing defects and the need for post-processing, such as rolling or heat treatment, reducing cost and waste. Developing a reliable and accurate numerical model promotes process robustness, stability and repeatability [86].

3.2. Qualification and certification

Qualification and certification are essential in commercializing additively manufactured products [87]. Qualification

refers to a process of evaluating a prototype design, product or process during its development or testing, whereas certification generally relates to satisfying the requirements of a certifying authority [88]. A reliable computational model can be part of process certification and part qualification, as is typically required for aerospace and medical applications. This approach may reduce the requirements for part quality and performance inspections. Detection of defects such as porosity requires destructive testing or the application of non-destructive techniques such as x-ray computed tomography (CT). The strong absorption of x-rays by metals, particularly steel and nickel-based alloys, limits the size of components amenable to x-ray CT scanning, depending on the x-ray energy. For example, the dimensions of a part are limited to about 40 mm for titanium and 20 mm for steel, based on 10% transmission of 240 keV x-rays. Measurement of residual stresses in larger components typically requires neutron scattering, which is only available at highly specialized facilities with its high cost limiting use to high-value parts.

It is relevant to note that a standard for arc welding simulation, ISO/TS 18166, has been developed [89]. Chen *et al* [88] noted the increasing focus on digital qualification and certification for metal AM. They suggested that this would reduce defects and improve product performance while requiring limited human intervention. The adoption of reliable numerical models is well suited to such a digital framework.

3.3. Process control and digital twins

A computational model can contribute to the control and management of the WAAM process by replicating important aspects of the process, helping to ensure that actions taken by the operator or an AI system during the production of a part are optimized. Gunasegaram *et al* [83] pointed out that AM is affected by the lack of reproducibility resulting from each component's particular processing route. This unstructured, unpredictable, and often disordered environment can be controlled by a digital twin incorporating a process model, improving reproducibility by exercising better control over the process.

Xia *et al* [90], Gunasegaram *et al* [83], and Phua *et al* [91] have emphasised the importance of digital twins in smart manufacturing [30, 83, 91]. Phua *et al* [91] proposed a digital twin hierarchy for AM segmented into four levels depending on the degree of communication with the physical process and automation. The base level, the implicit digital twin, entails modelling of the process to determine printing parameters. Subsequent levels (instantiated, interfaced and intelligent digital twins) require monitoring of the fabrication process and, in the latter two cases, feed-forward control of the process. In all cases, multi-physics models are a critical component. The requirements for models in a digital twin will be considered in section 6.4.1.

4. Numerical methods for WAAM modelling

Three prominent simulation tools, CFD, FEA and SPH, have been used to predict properties such as the flow field of the arc plasma and melt pool and the temperature distribution in the arc, melt pool and the surrounding metal in arc welding and WAAM. Magnetohydrodynamics (MHDs), which includes electromagnetic equations within CFDs, has been applied to predict arc plasma phenomena [21]. Table 1 highlights the differences in computational features and prediction accuracy, along with the respective strengths of CFD, MHD, FEA and SPH.

To model WAAM process using those approaches, the fundamental physics underlying each tool must be the same, even though the numerical approach and physics formulations may differ [10]. In this section, we introduce numerical modelling approaches, including these three methods, that can be adopted to model WAAM to investigate existing problems and process optimization. We also discuss how the simulation methods can capture the different phenomena occurring in WAAM.

4.1. Governing equations: fluid and heat transfer for arc and melt pool

The mathematical foundations for modelling the continuous phases associated with the WAAM process are based on principles of conservation of mass, momentum and energy. The continuity and the momentum equations, also known as the Navier–Stokes equations, are the governing equations for the liquid and the gaseous phases. The basic form of those equations for viscous compressible flow and including relevant source terms is as follows:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}) = 0 \tag{1}$$

$$\frac{\partial (\rho \boldsymbol{v})}{\partial t} + \nabla \cdot [\rho (\boldsymbol{v} - \boldsymbol{V}_{\mathrm{w}}) (v - V_{\mathrm{w}})] = \nabla \cdot \boldsymbol{\tau} - \nabla P + \boldsymbol{j} \times \boldsymbol{B} + \rho \boldsymbol{g} + S_{\mathrm{m}}. \tag{2}$$

Here ρ, $\boldsymbol{v}$, $\boldsymbol{V}_{\mathrm{w}}$, P, $\boldsymbol{\tau}$, $\boldsymbol{j}$, $\boldsymbol{B}$ and $\boldsymbol{g}$ denote the mass density, the flow velocity, the travel velocity (or welding velocity, which is the imposed velocity of the welding torch), the pressure, the stress tensor, the current density, the magnetic field induced by the current, and the acceleration due to gravity, respectively. Note that the mass density is usually assumed to be constant in the metal region, with buoyancy in the melt pool treated using a source term in equation (2), as described in section 4.5.3. The first term on the right-hand side of the momentum equation is the viscous stress term. The other terms are source terms representing, respectively, the pressure gradient, the Lorentz force and the gravity. S_{m} represents any additional momentum source terms, including the drag in the mushy zone, surface tension, buoyancy, recoil pressure and the rate of momentum transfer by droplets [10, 22].

An energy conservation equation is required to deal with heat transfer and phase change prediction occurring in the

Table 1. Comparison of CFD, MHD, FEA and SPH.

	Computational fluid dynamics (CFD)	Magneto-hydrodynamics (MHD)	Finite element analysis (FEA)	Smoothed particle hydrodynamics (SPH)
Media	Plasmas, gases, liquids, solids	Plasmas, gases, liquids, solids	Solids	Liquids, gases, solids
Mesh	Yes	Yes	Yes	No
Speed	Slow	Slow	Faster	Slow
Strengths	Fluid flow including plasmas, heat transfer	Fluid flow including plasmas, heat transfer, electromagnetic effects	Heat transfer, residual stress and distortion	Fluid flow, particularly with complex boundaries
Accuracy	High accuracy in fluid flow problems	High accuracy in electromagnetic effects on fluid flow problems	High accuracy in thermo-structural problems	High accuracy in free surface flows
Adaptability	Remeshing or mesh interface such as dynamic mesh is required for complex geometries and moving boundaries	Remeshing or mesh interface such as dynamic mesh is required for complex geometries and moving boundaries	Adaptable for complex geometries	Adaptable for complex geometries
Scalability	Efficient for large-scale problems; less efficient for multi-scale problems	Efficient for plasma flows	Efficient for structural analysis	Efficient for multiphase flows, particularly with liquid–solid interfaces

computational domain. It is often written in terms of the specific enthalpy h_s:

$$\frac{\partial(\rho h_s)}{\partial t} + \nabla\cdot[\rho(\boldsymbol{v}-\boldsymbol{V}_w)h_s] = \nabla\cdot\left(\frac{k}{c_p}\nabla h_s\right) + \frac{j^2}{\sigma} + \frac{5k_B}{2ec_p}j\cdot\nabla h_s - U + S_{ht}. \quad (3)$$

Here σ, k, c_p, k_B and e denote the electrical conductivity, thermal conductivity, specific heat at constant pressure, Boltzmann's constant and the charge on an electron, respectively. The first term on the right-hand side represents thermal conduction. The other terms on the right-hand side are all source terms, with the second term representing resistive heating. The third and fourth terms denote the energy transfer occurring due to the electron flow and the radiative emission, respectively, which can be omitted outside the arc plasma as these are negligible; U is the net radiative emission coefficient. S_{ht} represents any additional heat transfer source terms at the boundaries between the arc and the metal, and the solid and liquid metal; these are considered in section 4.6.1. If metal vapour in the arc is modelled, an additional equation for the conservation of metal vapour mass fraction is required, together with an additional term in the energy conservation equation to treat mixing of the metal vapour and the shielding gas [92], as discussed in section 4.6.2. It is important to note that the assumption of local thermodynamic equilibrium (LTE) in the plasma region is used in equation (3). LTE allows the plasma to be described by a single temperature, and is a reasonable assumption in welding arcs, except in the regions close to the electrodes and the fringes of the arc [93]. Deviations from LTE near the electrodes are discussed in section 4.6.1; the deviations in the fringes of the arc have negligible influence in WAAM since the energy transfer occurs predominantly through the central regions of the arc.

Modelling the flow of current requires a current continuity equation:

$$\nabla\cdot\boldsymbol{j} = -\nabla\cdot(\sigma\nabla\varphi) = 0 \quad (4)$$

where φ is the electric potential. The magnetic field strength $\boldsymbol{B}$, required in equation (2), can be obtained from the magnetic potential $\boldsymbol{A}$ using $B = \nabla\times A$, which requires the solution of the equation:

$$\nabla\times\boldsymbol{A} = -\mu_0 j. \quad (5)$$

Alternatively, $\boldsymbol{B}$ can be calculated directly using the Biot–Savart equation, although this is computationally intensive [94]. Note again that the combination of CFD and electromagnetic equations is often referred to as MHD.

4.2. SPH discretisation of heat and fluid flow governing equations

A detailed mathematical description of SPH modelling can be found in, for example [95–98]. The continuity, momentum, and energy equations are solved in the Lagrangian frame of reference by converting them into a SPH spatial discretization [23]. The SPH spatial derivatives of interpolants are calculated over a local region defined by the interpolation kernel

$W(\boldsymbol{r}-\boldsymbol{r}_b,h)$. The interpolated value at position $\boldsymbol{r}$ of a function A is given by the sum over particles within the kernel radius of $2h$:

$$A(\boldsymbol{r}) = \sum_b m_b \frac{A_b}{\rho_b} W(\boldsymbol{r}-\boldsymbol{r}_b,h). \tag{6}$$

Here m_b, ρ_b, $\boldsymbol{r}_b$ and h denote the mass, density and position of particle b and the interpolation length, respectively. The term $W(r-\boldsymbol{r}_b,h)$ describes the smoothing kernel with radius $2h$, which is required to be at least twice differentiable, and is usually selected as an approximation of the shape of a Gaussian function [97, 98].

The differentiation of the interpolation equation (5) gives the gradient of the function A_b at position r [97, 98]:

$$\nabla A(\boldsymbol{r}) = \sum_b m_b \frac{A_b}{\rho_b} \nabla W(\boldsymbol{r}-\boldsymbol{r}_b,h). \tag{7}$$

This enables the conversion of the governing equations for continuous phases, which are in the form of parabolic partial differential equations (PDEs), to ordinary differential equations for the particle motion and the rates of change of their properties in the Lagrangian frame [98]. The continuity equation in the form of the SPH discretization suitable for use in free surface flows can be expressed as:

$$\frac{d\rho_a}{\mathrm{d}t} = \sum_b m_b (\boldsymbol{v}_a - \boldsymbol{v}_b)\cdot\nabla W_{ab} \tag{8}$$

where ρ_a and v_a respectively denote the density and velocity of particle a and v_b denotes the velocity of particle b. Similarly, the basic fluid flow momentum equation also can be written in the spatially discretised form as:

$$\frac{\mathrm{d}\boldsymbol{v}_a}{\mathrm{d}t} = \sum_b m_b \left(\frac{\boldsymbol{\sigma}_a}{{\rho_a}^2} + \frac{\boldsymbol{\sigma}_b}{{\rho_b}^2} + \boldsymbol{\Pi}_{ab}\boldsymbol{I}\right)\cdot\nabla_a W_{ab} + \boldsymbol{g}_a \tag{9}$$

where $\boldsymbol{\sigma}_a$ and $\boldsymbol{\sigma}_b$ denote the stress tensors of particles a and b. Π_{ab} denotes a viscosity used to generate a shear and can be calculated as presented in, for example [20, 99]. SPH forms of other momentum terms (such as surface tension, buoyancy etc) can be added to the basic form of the SPH momentum equation (8).

Cleary [100] provided a derived form of the energy equation for SPH:

$$\frac{\mathrm{d}h_a}{\mathrm{d}t} = \sum_b \frac{4m_b}{\rho_a\rho_b}\frac{k_a k_b}{k_a+k_b} T_{ab}\frac{r_{ab}\cdot\nabla_a W_{ab}}{{r^2}_{ab}+\delta^2} + \Phi_a \tag{10}$$

where h_a, k_a, k_b and δ respectively denote the specific enthalpy of particle a, the thermal conductivities of particles a and b, and a small number preventing a zero denominator, while T_{ab} and r_{ab} are $T_a - T_b$ and $r_a - r_b$, respectively. Φ_a includes the rate of cooling caused by convection or radiation when the particle a is a free surface in a solid or fluid state [20], and all extra heat source terms can be consolidated for WAAM. The surface flux component can be determined as follows [100]:

$$\Phi_a = \frac{1}{\rho_a}\sum_b 2m_b k_b T_{ab} r_{ab}\cdot n_a \frac{r_{ab}\cdot\nabla_a W_{ab}}{{r^2}_{ab}+\delta^2}. \tag{11}$$

Here n_a is the unit normal to particle a. Also, the specific enthalpy, taking into account a phase change and the temperature dependence of the specific heat, is defined as [20, 99]:

$$h_s = \int_T^0 c_p(\theta)\,\mathrm{d}\theta + L f_l(T) \tag{12}$$

where L and $f_l(T)$ denote the latent heat of fusion and the liquid volume fraction at temperature T, respectively.

4.3. Governing equations: FEA for thermal and residual stress modelling

CFD is typically applied to WAAM by those interested in the melt pool created on the substrate, the interaction of the droplet with the melt pool, the shape of the solidified droplet, and, in some cases, the arc plasma and its interactions with the droplet and the melt pool. Conversely, where the research or engineering focus is solely on the larger scale thermal and deformation behaviour of the complete solid part then simplifications need to be made to the physics to allow for the large scale of calculation needed. With appropriate simplifying assumptions about the fluid flow, the WAAM process can be modelled allowing residual stresses and distortion prediction for entire parts. The broadly preferred numerical method for solving this type of problem has been FEA.

When such a modelling approach is applied to WAAM or arc welding, the arc plasma is not modelled, and the melt pool is treated as a solid. Therefore, the Navier–Stokes equations are not used, with some modellers using an arbitrarily high value (for example, a factor of 10 times higher) for thermal conductivity to account for convection-induced equilibration of temperature gradients (e.g. [24, 101]). FEA is typically used to predict the stresses that develop and accumulate in the part because of the thermal and mechanical loads that result in a specific zone of the material experiencing resistance to expansion or contraction from its surroundings. Thermal loads may take the form of steep macroscopic temperature gradients (owing to the previously deposited layers being at a much lower temperature and resisting the contraction of the recent layers), the localized and recurrent nature of heat input resulting in non-uniform and repetitive heating and cooling around the melt pool (where the melt pool's ability to shrink as it cools is opposed by an expanding substrate which has just received a pulse of heat from the melt pool), and the phase transformation process (which results in a large solidification shrinkage). The mechanical load results from the constraint presented by the clamped substrate to the thermally caused movements of the part. Much of these stresses remain as 'residual stresses' after the WAAM process and are only partially relieved when the newly built part is released from its clamps. During this

stress-relieving process, the part plastically deforms and can become unusable if the deformation is excessive. Indeed, distortion is a major problem that must be carefully considered in planning WAAM builds.

FEA is the process of carrying out analyses using the FEM. FEM [102] is the most common choice for modelling the solid's behaviour in AM based on solid mechanics principles, providing a platform for predicting the residual stress and distortion. Predicting the residual stresses requires calculation of the thermal history (the time-dependence of the temperature distribution) of the substrate. Since Joule heating of the substrate can be significant, the electric current distribution is also calculated in some cases.

Essentially, the FEA formulation for the reduced mechanical physics as applied to WAAM involves the solution of simultaneous equations based on the need for mechanical equilibrium (i.e. no motion) and Hooke's law:

$$\boldsymbol{F} = k_s \boldsymbol{u} \tag{13}$$

where F denotes the external force vector, k_s the stiffness, and u the nodal displacement vector. In this equation, applicable to linear problems where k_s is independent of $\boldsymbol{F}$ and $\boldsymbol{u}$, $\boldsymbol{F}$ is calculated from applied loads and constraints to movement, and k_s is calculated from material properties. The equation is solved to obtain u, corresponding to the simulated system's response to the applied loads. In transient simulations, which are required to simulate WAAM, F and k_s are updated at each time step before the equation is solved for u at that instant. Since the solution for each node must be found, matrix algebra is used to find $\{\boldsymbol{u}\} = \{\boldsymbol{F}\} / [\boldsymbol{k_s}]$ through matrix inversion. If there are boundary conditions involved, e.g. the displacements at some nodes being required to be zero (or any other value), these are taken into consideration when assembling $[\boldsymbol{u}]$. The same methodology is repeated equivalently for other physics. For example, in thermal and electrical physics, the system response $\boldsymbol{u}$ will be equivalent to temperature and electrical current, respectively. Similarly, the driving force $\boldsymbol{F}$ will be equivalent to heat flux and voltage. In mechanical physics, the values of $\boldsymbol{u}$ calculated at the nodes of an element provide the deformation pattern of that element. The $\boldsymbol{u}$ values are then converted to strain ε at each node, e.g. in the x-direction:

$$\varepsilon_x = \frac{\partial u}{\partial x} \tag{14}$$

which formally states that normal strain is the rate of change of u in the x-direction with respect to position. Once the strain is calculated, the normal stress profile σ_x within the element may be calculated by connecting ε_x and ε_y with the elastic modulus E and Poisson ratio ν, using the following relationships (for a two-dimensional stress state):

$$\sigma_x = E\frac{(\varepsilon_x + \nu\varepsilon_y)}{(1-\nu^2)} \tag{15}$$

$$\sigma_\psi = E\frac{(\varepsilon_y + \nu\varepsilon_x)}{(1-\nu^2)} \tag{16}$$

and the shear stress state:

$$\tau_{xy} = E\frac{\gamma_{\text{bulk}_{xy}}}{(1+2\nu)} \tag{17}$$

where γ_{bulk} is the bulk modulus.

For thermal physics, the heat flow within a solid is calculated using the Fourier heat conduction equation in one direction:

$$q_x = -kA_x\frac{\partial T}{\partial x} \tag{18}$$

where q is the thermal load and A_x is the area through which the heat flows. The above equation formally states that the heat flux q_x is proportional to the temperature gradient in the x-direction. Equivalent to mechanical physics, temperatures at the various nodes are solved for simultaneously using a matrix formulation. For a steady-state condition, the net rate of heat flow into any element is zero:

$$[K]\{T\} = \{q\} \tag{19}$$

where $[K]$ is the conductivity matrix, $\{T\}$ is the nodal temperature matrix and $\{q\}$ is the thermal load matrix. For unsteady state problems found in processes such as WAAM, the equation becomes:

$$[C]\left\{\dot{T}\right\} + [K]T = \{q\} \tag{20}$$

where $[C]$ is the heat capacity matrix and $\left\{\dot{T}\right\}$ denotes the time rate of change of temperature at the nodes. The additional term in this equation accounts for energy storage or loss in an element (i.e. net heat flow into the element is non-zero).

We have only provided a basic overview of the FEA methods relevant to WAAM. In-depth coverage may be found elsewhere in various textbooks; for example, Liu and Quek [102].

4.4. Solid-liquid interface

In metal alloys, the interface between solid–liquid phases extends from the solidus temperature T_{sol} to the liquidus temperature T_{liq}. The intermediate region, which is a mixture of solid and liquid phases, is known as the mushy zone. The liquid fraction is usually assumed to increase linearly with temperature:

$$f_{\text{l}} = \begin{cases} 1 & T \geqslant T_{\text{liq}} \\ \frac{T-T_{\text{sol}}}{T_{\text{liq}}-T_{\text{sol}}} & T_{\text{sol}} < T < T_{\text{liq}} \\ 0 & T \leqslant T_{\text{sol}} \end{cases}. \tag{21}$$

The latent heat of fusion is similarly applied linearly from the solidus to liquidus temperature, as shown in equation (21). In some implementations, the enthalpy h_s in the energy conservation equation (3) is treated as the sensible enthalpy (i.e. the enthalpy excluding the latent heat of fusion), and the latent heat of fusion is included as a source term, following the methods of Voller and Prakash [103].

Voller and Prakash further hypothesized that the mushy zone can be considered as a porous medium [103], where fluid

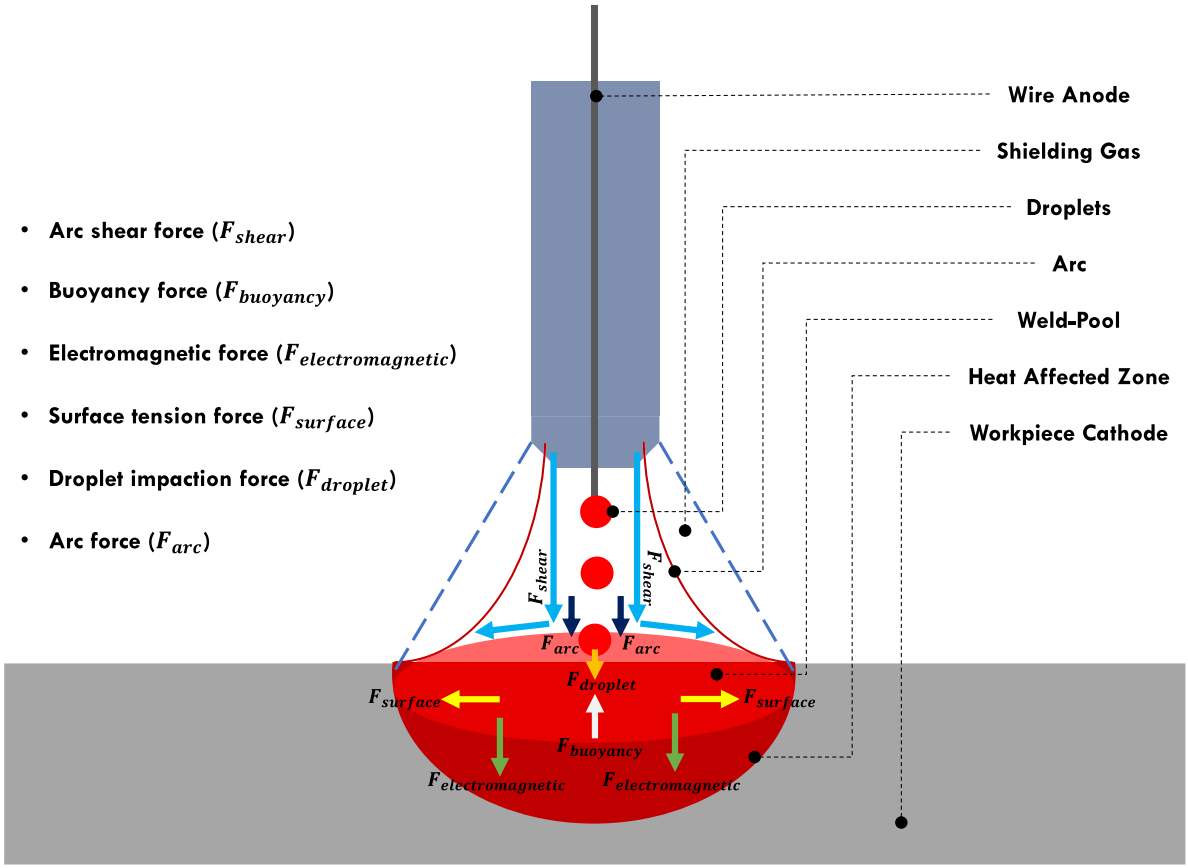


Figure 3. Schematic diagram showing forces acting on the melt pool. Reproduced from [48]. © IOP Publishing Ltd All rights reserved.

flow within the porous medium can be described by the Darcy law. Brent *et al* [104] defined the porosity function, $A(f_l)$, which is commonly used as a source term in the momentum equation to represent fluid flow through the porous medium, as

$$A\left(f_l\right) = \frac{A_{\text{mush}}\left(1-f_l\right)^2}{{f_l}^3+\varepsilon} \tag{22}$$

where A_{mush} and ε denote the mushy zone constant and a small computational constant preventing zero in the denominator, respectively. Further information regarding the liquid fraction can be found in the work of Fadl and Eame [105].

4.5. Forces acting on the melt pool

Several physical factors affecting the shape and properties of a deposited material must be considered in fluid-focussed WAAM modelling. Figure 3 is a schematic of the forces acting on the melt pool, which include the shear force and pressure applied by the arc, the droplet impact force, buoyancy, electromagnetic force and surface tension. These forces, which contribute significantly to the momentum and heat transfer, are discussed below.

4.5.1. Surface tension. Surface tension affects interfaces between the arc plasma and the molten wire tip, the metal droplets and the melt pool. The surface tension force can be divided into components normal and tangential to the surface as follows [106]:

$$F_\gamma = \gamma\kappa\widehat{\boldsymbol{n}} + \left[\nabla\gamma - \widehat{\boldsymbol{n}}\left(\widehat{\boldsymbol{n}}\cdot\nabla\right)\gamma\right]. \tag{23}$$

Here γ, κ, $\widehat{n}$ are, respectively, surface tension, the surface curvature and the surface unit normal vector. The first term influences the curvature of the melt pool surface and the shape of any pores within the liquid [10]. The second term describes the Marangoni effect and acts tangentially to the surface, driving flow due to temperature and composition gradients at the melt pool surface. The mathematical expression for the surface tension γ, including the effect of surfactants is [107]:

$$\gamma = \gamma_m^o + \left(\frac{\partial\gamma}{\partial T}\right)^o\left(T-T_m\right) - RT\Gamma_s\, ln\left[1+\Psi\, a_s exp\left(-\frac{\Delta H^o}{RT}\right)\right] \tag{24}$$

where $\gamma_{\text{m}}^{\text{o}}$, T_{m} and $(\partial\gamma/\partial T)^o$ are the surface tension at the melting temperature, the melting temperature, and the temperature coefficient of the surface tension of pure molten metal, respectively. Also, R, Γ_{s}, Ψ, a_{s} and ΔH^{o}denote, respectively, the ideal gas constant, the adsorption at saturation point, the entropy factor, the activity of the solute and the standard heat of adsorption. Most authors only consider the temperature gradient term.

4.5.2. Lorentz force. The Lorentz force, which is given by the $\boldsymbol{j} \times \boldsymbol{B}$ term in the momentum equation, is induced by the current flow through the arc and metal. It is the dominant force driving flow in the arc but also influences flow in the melt pool. If the arc plasma is included in the computational domain, the current density and magnetic field are obtained directly. If not, then approximate expressions are required. If electromagnetic equations are included in the model (e.g. in an MHD model), the vector fields of the current density and the magnetic field can be set as boundary conditions for the sub-region of the melt pool at the metal–arc interface [108]. Otherwise, electromagnetic force models are used. Two analytical electromagnetic force models, developed by Kou and Sun [109] and Tsao and Wu [110], respectively, are widely adopted for numerical simulations [108]. Bai *et al* [111] described the multiaxial electromagnetic forces per unit area acting in the melt pool, based on the approach of Tsao and Wu [110], as:

$$F_x = -\frac{\mu_m I^2}{4\pi^2 \sigma_j} \exp\left(-\frac{{r_{eff}}^2}{2{\sigma_j}^2}\right)\left[1-\exp\left(\frac{{r_{eff}}^2}{2{\sigma_j}^2}\right)\right]\left(1-\frac{z}{L_0}\right)^2 \times \left(\frac{x-x_a-V_W t}{r}\right) \tag{25}$$

$$F_y = -\frac{\mu_m I^2}{4\pi^2 \sigma_j} \exp\left(-\frac{{r_{eff}}^2}{2{\sigma_j}^2}\right)\left[1-\exp\left(\frac{{r_{eff}}^2}{2{\sigma_j}^2}\right)\right]\left(1-\frac{z}{L_0}\right)^2 \times \left(\frac{y-y_a}{r}\right) \tag{26}$$

$$F_z = -\frac{\mu_m I^2}{4\pi^2 \sigma_j} \exp\left(-\frac{{r_{\mathrm{eff}}}^2}{2{\sigma_j}^2}\right)\left(1-\frac{z-z_a}{L_0}\right). \tag{27}$$

Here, μ_m, σ_j, L_0 and r_eff denote, respectively, the substrate permeability, the effective radius of the current density distribution and the substrate thickness and the radius, which is given by $r_\mathrm{eff} = \sqrt{(x-x_a-V_W t)^2 + (y-y_a)^2}$; x_a, y_a and z_a are the coordinates of the arc position. Norrish *et al* [8] expressed the electromagnetic force in a simpler form, assuming axisymmetric and that the current flow diverges uniformly. As the electromagnetic force acting perpendicular to the surface of the melt pool is much larger than the radial component, the electromagnetic force can, therefore, be approximated by its axial component, $F_{\mathrm{em},z}$:

$$F_{\mathrm{em},z} = \frac{\mu_\mathrm{m} I^2}{4\pi} \ln\left[\frac{r_\mathrm{eff}(z)}{r_\mathrm{eff}(0)}\right] \tag{28}$$

where the logarithm term is a geometric factor that denotes the axial variation.

It has been observed that the electromagnetic force can significantly affect the fluid flow in the melt pool [112, 113] and influence both the melting and solidification of the metal [114–116].

4.5.3. Buoyancy force. The buoyancy force, which corresponds to the gravitational force term in equation (2), arises from variations in the density of the melt pool due to temperature variation [117]. The buoyancy force acts upwards while the electromagnetic force acts downwards and the drag force induces outwards flow. The buoyancy force, F_b, is usually calculated by the Boussinesq approximation [118]:

$$F_\mathrm{b} = -\rho\beta g\,(T - T_\mathrm{L}) \tag{29}$$

where β and $T - T_\mathrm{L}$ denote, respectively, the thermal expansion coefficient and the difference between the local temperature and a reference temperature, usually the liquidus temperature [10]. Generally, the buoyancy force is small in comparison to other forces acting on the melt pool, in particular, the surface tension and electromagnetic forces [119].

4.5.4. Arc shear force. The arc shear force is a drag force associated with the shear stress that the arc plasma flow applies to the surface of the melt pool, which typically induces a radial outflow across the surface [120]. The arc shear force is a function of the shear stress and the area over which the arc shear stress is distributed. The arc shear stress, τ_a, can be expressed as:

$$\tau_a = \mu_\mathrm{p} \frac{\mathrm{d}v_\mathrm{p}}{\mathrm{d}z_\mathrm{perp}} \tag{30}$$

where μ_p, v_p and z_perp denote the dynamic viscosity of the arc plasma, the arc plasma velocity component parallel to the melt-pool surface and the coordinate perpendicular to the melt-pool surface, respectively [8, 121, 122]. An arc plasma model is required to calculate μ_p and v_p, and the arc shear force is usually ignored if the arc is not modelled. Tanaka *et al* [123] showed that the shear force for an argon arc made a substantial contribution to the predicted outward radial flow near the melt pool surface, so neglecting the shear force can lead to significant inaccuracy. Campbell *et al* [120] compared the shear stress distribution along the radius from the centre of the droplet deposition with different shielding gases. The peak shear stress occurred approximately 15% from the centre of the droplet deposition and was found to be higher for helium than argon shielding gas.

4.5.5. Force due to arc pressure. The pressure from the arc is another significant component acting on the free surface of the melt pool. If the arc is modelled, the pressure is obtained from the arc velocity distribution. If not, approximate expressions are often used. Bai *et al* [111] approximated the arc pressure by a Gaussian density distribution, expressed as:

$$P_\mathrm{arc} = \frac{F_\mathrm{arc}}{2\pi {\sigma_\mathrm{p}}^2} \exp\left(-\frac{{x_a}^2 + {y_a}^2}{2{\sigma_\mathrm{p}}^2}\right) \tag{31}$$

where σ_p determines the radius over which the arc pressure acts and F_arc is the force exerted by the arc. Corradi *et al* [124] expressed the arc plasma force as:

$$F_{arc} = \frac{\mu_0 I^2}{8\pi}\left[2\ln\left(\frac{R_{anode}}{R_{cathode}}\right) + 1\right] \tag{32}$$

where μ_0, R_{anode} and R_{cathode} are the permeability of free space, and the radius of the arc at the anode and cathode, respectively.

4.5.6. Force due to droplet impact. The momentum transferred to the melt pool by the molten metal droplet from the feed wire is a primary factor affecting the shape of the melt pool. The impact force due to the droplet can be determined from the impact kinetic energy of the droplet, which can be expressed as:

$$KE_{\text{droplet}} = \frac{2}{3}\rho\pi {R_{\text{droplet}}}^3 |v_{\text{droplet}}|^2 \tag{33}$$

where R_{droplet} and v_{droplet} are the radius and the flight velocity of the droplet, respectively. Feng *et al* [125] observed the dependence of the impact behaviour of the droplet on the deposition (or welding) speed, finding a critical deposition speed corresponding to the transition between the droplets depositing inside or outside the melt pool. It is also important to note that the width of the deposited material decreases if the deposition velocity increases, showing that the droplet impact can contribute significantly to the shape and dimension of the melt pool as well as the formation of the spatter.

4.6. Heat transfer at interfaces

4.6.1. Arc-metal interface. In computational models that do not include the arc plasma in the computational domain, which include all FEA models, an analytic expression is generally used to approximate the heat transferred from the arc to the workpiece.

Commercial FEA codes offer several options to introduce the heat load Q_{T} directly onto the substrate. The common choices include the definition of a volume flux on the substrate elements or a surface flux on the top surfaces of substrate elements.

The volume definitions include the two-dimensional shape of the footprint on the surface of the substrate and the 3D shape of the penetration volume under the surface (i.e. the melt pool shape). A cylindrical shape would typically be assumed when employing a laser as the source. When using an arc, as in WAAM, Goldak's double ellipsoidal shape [126] or a twin double ellipsoidal shape [127] are frequent choices. These shapes, shown in figure 4, represent the melt pool's shape attributable to the arc's heat penetration. Additionally, the elongated nature of the shapes accounts for the arc movement. An overwhelming majority of works in the literature have used the double ellipsoidal shape, which is expressed as:

$$S_{ht\,\zeta}(x,y,z,t) = \frac{6\sqrt{3}f_\zeta Q_T}{abc_\zeta\pi\sqrt{\pi}}\exp\left[\frac{-3(x-V_W t)^2}{c_\zeta^2}\right] \times \exp\left(\frac{-3y^2}{a^2}\right)\exp\left(\frac{-3yz^2}{b^2}\right) \tag{34}$$

where ζ takes the value f or r, with f_{f} and f_{r} representing the front and rear distribution fractions, respectively, and a, b, c_{f} and c_{r} are the dimensions shown in figure 4(a). The twin double ellipsoidal shape is used when 'the cross-section of the weld shows a finger or a deeper penetrated section', as shown in figure 4(b) [127].

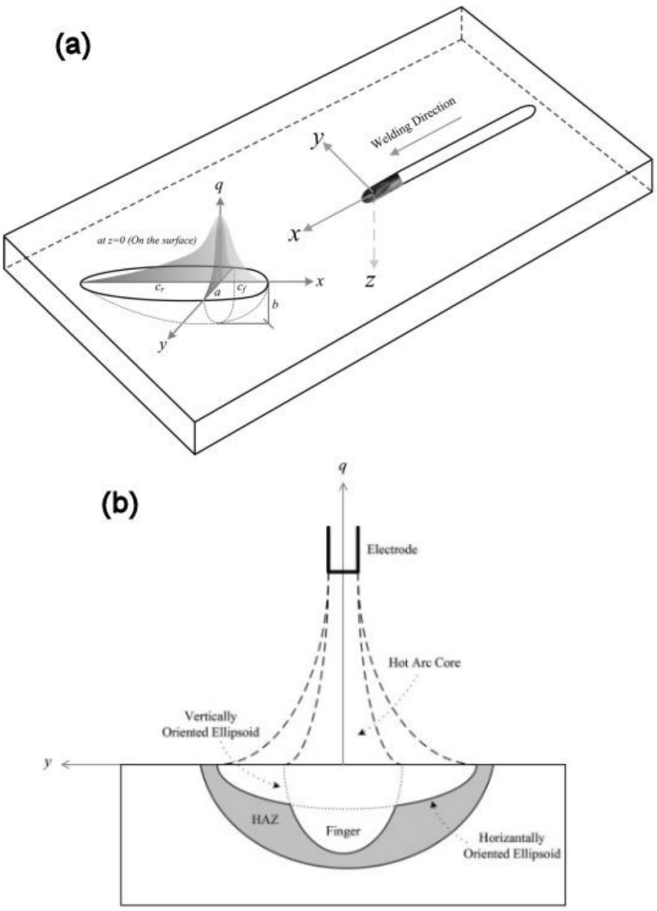


Figure 4. The assumed shapes of the melt pool in the substrate (which correspond to the 'heat load' in FEA) when an arc is used in WAAM to provide the heat. The Goldak double ellipsoidal shape (a) or the twin double ellipsoidal shape (b) may be used depending on the heat penetration mechanism encountered. Reprinted from [127], Copyright (2012), with permission from Elsevier.

The heat transferred to the substrate Q_{T} may be divided into the heat quantity $Q_{\text{substrate}}$ directly transferred to the substrate from the arc and the heat Q_{droplet} contained within the droplets, making $Q_{\text{T}} = Q_{\text{substrate}} + Q_{\text{droplet}}$. In this case, the surface flux option would be used for the arc heat:

$$S_{ht\,\zeta}(x,y,t) = \frac{6\sqrt{3}f_\zeta Q_S}{ac_\zeta\Delta l\pi\sqrt{\pi}}\exp\left[\frac{-3(x-V_W t)^2}{c_\zeta^2}\right]\exp\left(\frac{-3y^2}{a^2}\right) \times + \frac{S_{droplet}}{\Delta l} \tag{35}$$

where $S_{droplet} = Q_{droplet}/(\pi R_{droplet}^2)$ is the power per unit area transferred by the droplet, R_{droplet}^2 being the droplet radius. Δl is a characteristic length scale, introduced to give the source term the correct dimensions. It is usually set to the length of the mesh element perpendicular to the electrode surface, or a

similar length in a meshless scheme, and represents the length over which the power transferred by the droplet is dissipated.

A critical point is that the user needs to nominate the dimensions *a, b,* c_{f} and c_{r} of the ellipsoidal shapes and the arc efficiency ε (since $Q_T = \varepsilon VI$, where V is the arc voltage, respectively). If the droplet heat is treated separately, the attribution of Q_{T} between $Q_{\mathrm{substrate}}$ and Q_{droplet} also must be determined. A complicating factor in WAAM is that the dimension and efficiency factors change as layers are deposited because the arc interacts differently with the initial substrate and the top of several layers. The required dimensions and efficiency are typically determined 'after the event' using a trial-and-error approach. For example, the thermal history predicted by the model can be compared with measurements performed using thermocouples in the substrate. This approach is widely used in arc welding but is less reliable in WAAM because the thermocouples become progressively more distant from the heat source as layers are deposited. Alternatively, the predicted melt pool dimensions can be compared with the fusion zone visible in a cross-section; ideally, cross-sections would be examined after the deposition of each layer.

Although ellipsoidal shapes are commonly used, other shapes have also been assumed by modellers. For instance, some researchers, e.g. [128, 129], have idealized the ellipsoidal shape into a cuboid for convenience of definition in a Cartesian system. Similarly, Ding *et al* [101] compared the computational efficiency relating to three heat source shapes (double ellipsoidal, hemispherical with a Gaussian profile, and a cuboid volume heat source). It would be useful to determine how much influence the assumed heat source shape exerts on predicted temperature profiles. The exercise is, however, likely to be complicated by inherent approximations, such as the accuracy of the material properties and boundary conditions. In FEA, the melt-pool fluid dynamics is not considered while the changing shapes of the layers is also not accommodated since they slump during solidification. As a result, the final solid shape is frequently approximated. It, therefore, becomes difficult to apportion the total error (i.e. the difference between predicted and experimental temperature histories) between these contributors and the assumption of the heat source shape. However, if only the heat source shape is changed between simulations, a rough estimate of its influence may still be obtained.

When the arc is modelled, it is possible to calculate the heat flux S_{ht} directly. However, the approach used must be carefully considered, since heat transfer at the interface between the arc plasma and the electrodes is a complicated topic that is the subject of continuing research. While LTE is a reasonable assumption in most of the arc plasma, deviations from thermal equilibrium between the electrons and heavy species in the plasma occur within about 0.1 mm of the electrode, and deviations from charge neutrality occur within about 1 μm of the electrodes [92]. Sophisticated techniques have been developed to treat these regions (e.g. Benilov *et al* [130], Baeva [131]); however, these are computationally intensive and have not been integrated into arc welding or WAAM simulations to date. Simplified treatments, in which the non-equilibrium regions are not resolved, are widely used [132]. In such treatments, additional heat transfer terms are included at the arc-electrode interface. These account for the heat transferred to the electrode due to sheath effects and by the droplets, the cooling of the cathode due to the energy required to remove electrons from the metal, the heating of the anode when the electrons return to the metal in arc, and the cooling due to blackbody radiation and metal vapour production. A typical expression for the heat transferred to the cathode is:

$$S_{\mathrm{ht}} = \left[j_{\mathrm{e}}\left(V_{\mathrm{s}} - \varphi_{\mathrm{cathode}}\right) + S_{\mathrm{droplet}} - S_{\mathrm{vap}} - S_{\mathrm{bb}}\right]/\Delta l \qquad (36)$$

where j_{e} is the current density at the electrode surface, V_{s} is the sheath voltage, $\varphi_{\mathrm{cathode}}$ is the work function of the cathode material, $S_{\mathrm{vap}} = J_{\mathrm{vap}}H_{\mathrm{vap}}$ is the cooling due to the latent heat of vaporisation H_{vap} and J_{vap} is the metal vapour flux from the electrode surface (see equation (26)), and S_{bb} is the blackbody cooling. At the wire anode, the first term is replaced by j_{e} φ_{anode}, where φ_{anode} is the work function of the anode material, and the droplet energy is negative since the droplets remove energy from the wire.

4.6.2. Metal vapour production.

The shape and dimension of the melt pool can also be significantly affected by the metal that evaporates from the liquid surfaces of the melt pool, the wire tip and the droplets. Vaporization cools the melt pool because of the latent heat required for the phase transition of the molten metal. Further, the metal vapour influences the arc properties [42] by increasing the radiative emission from the arc. Detailed mechanisms and observations of the effect of metal vapour on the arc and melt pool can be found in the literature [133–138]. The rate of vaporization depends on the metal vapour pressure, P_{vap}, which is usually calculated from the Clausius–Clapeyron equation:

$$P_{\mathrm{vap}} = P_{\mathrm{atm}} \exp\left[\frac{-H_{\mathrm{vap}}}{R_{\mathrm{g}}}\left(\frac{1}{T_{\mathrm{l}}} - \frac{1}{T_{\mathrm{b}}}\right)\right] \qquad (37)$$

where P_{atm}, H_{vap}, R_{g}, T_{l} and T_{b} denote atmospheric pressure, the molar heat of vaporisation, the ideal gas constant, the liquid metal temperature and the boiling temperature of the metal at atmospheric pressure, respectively. The metal vapour flux is usually calculated using the Hertz–Langmuir equation, considering the influence of the metal vapour already in the plasma, which slows evaporation. A typical expression is:

$$J_{\mathrm{vap}} = \frac{16}{9}\left(\frac{m_{\mathrm{M}}}{2\pi k_{\mathrm{B}}}\right)^{1/2}\frac{P_{\mathrm{vap}}}{T_{\mathrm{l}}^{1/2}}\left(\frac{P_{\mathrm{vap}} - P_{\infty}}{P_{\mathrm{vap}}}\right) \qquad (38)$$

where m_{M} is the mass of a metal atom and P_{∞} is the partial pressure of the metal vapour in the plasma at a small distance from the metal [22].

4.7. CFD modelling of WAAM

CFD models a continuous fluid phase, such as a liquid, gas, plasma or highly dense particles, in a Eulerian frame of

reference using either the finite volume method (FVM), FEM or finite difference method (FDM). The FVM is the most common method for fluid flow modelling, followed by FEM and FDM, with the choice depending on the desired complexity of mathematical implementation and level of accuracy.

CFD simulations can be either steady-state or transient. In WAAM, predictions of bead geometry, the cross-section of the deposited layer [139] and the fusion zone and heat-affected zone profiles [22] are particularly important for comparison with the physical domain. They can be predicted by a steady-state simulation under certain circumstances. For example, a steady-state simulation is possible for the deposition of the initial layer and subsequent layers when the build is allowed to reach a uniform temperature before deposition starts, in regions where the WAAM parameters are constant in time. When these conditions are not satisfied, a time-dependent calculation is required. In many cases, inherently time-dependent processes, such as droplet detachment and transfer, are modelled. Analysis of properties such as the variation of temperature with the macroscopic morphology of the deposition layer, which may be useful for investigating defect formation, requires a time-dependent model [139].

CFD modelling of the melt pool typically assumes that the flow field in the melt pool is in the laminar flow regime. Most CFD models of WAAM only consider flow in the melt pool, with the arc represented by a heat flux source term, and sometimes a momentum or pressure source term and an electromagnetic force source term, applied to the melt pool. Standard expressions for the heat flux, arc pressure and electromagnetic force are given in sections 4.6.1, 4.5.5 and 4.5.2, respectively. If the flow of electric current in the metal is considered, a current density source term is required instead of an electromagnetic force source term.

The fluid flow in the melt pool is assumed to be incompressible Newtonian flow [10], so the density ρ in equations (1)–(3) is constant. The Boussinesq approximation (equation (29)) is usually adopted to treat the variation of density with temperature. In the arc, the flow needs to be treated as compressible because of the large range of temperatures—typically from 300 K to at least 10 000 K—and consequently, densities. The variation of density and other thermophysical properties with temperature is usually pre-calculated using methods of statistical physics and thermodynamics [92] and parameterised by an analytic equation or included in a look-up table. The flow in plasmas of industrial interest, such as the plasma jets used for plasma spraying and other applications, can be turbulent and supersonic. However, in welding arcs, the flow speeds are less the 20% of the speed of sound, which increases with temperature, and the flow is laminar. This allows significant simplification: no additional equations are required to treat turbulent flow, special numerical methods to handle large pressure variations are not needed, and the pressure dependence of the thermophysical properties can be neglected.

4.8. SPH modelling of WAAM

SPH was initially developed to solve astrophysics problems in the 1970s [140] and has since been applied to simulations of phenomena including incompressible flows with boundaries [141], forming and solidification [142, 143], as well as complex fluid interactions and fracture [144, 145]. Cleary *et al* first implemented SPH techniques to simulate the metal flow [146], and subsequently these authors included heat transfer and realistic part geometries for their SPH code [147]. In 2006, Cleary and his colleagues expanded the applicability of their SPH model to the manufacturing sector by extending its capability to solid process modelling, including forging and extrusion, for high pressure die casting [148] and metal forming [149]. SPH also has been extensively used to simulate various industrial single- or multi-phase Newtonian fluids as well as non-Newtonian fluid flow problems including complex rheological and turbulent flows [95].

Monaghan comprehensively reviewed the underlying theory of SPH together with its potential applications, proposing that SPH could be expanded to treat diverse industrial problems [98]. He noted five advantageous features of SPH: an exact treatment of pure advection, usage of multi-materials with segmented description of each material, bridging the gap between the continuum and fragmentation in a natural way, position and time-dependent resolution, computational advantage over problems involving splashing of liquid phase, and finally, comprehensive inclusion of underlying multi-scale physics. Such features make SPH particularly suitable for problems associated with heat conduction and material diffusion, viscous flows, shock and rarefaction problems, fluid-structure interactions, the Stefan problem, multiphase flows and even turbulent flows (Cleary *et al* addressed the validity of SPH for high-speed turbulent flow simulation—the spatial straining technique used for the large eddy simulation in Eulerian models can be coordinated with the spatial interpolation used for SPH [95]).

Treatment of solidification during the casting process was first demonstrated in SPH by Cleary to simulate low-pressure [150] and high-pressure die casting [151]. These models have full internal heat conduction with suitable surface temperature boundary conditions, so that solidification can be controlled by the temperature. Further, SPH was applied to arc welding by Das and Cleary [20, 99], demonstrating plastic deformation of the filler metal, derived from melting of the wire electrode, during the arc welding by using the von Mises plasticity model. The model considered a single-phase material, i.e. both the filler and the workpiece metals were considered as solid. The filler material was assumed to be a plastic material having a temperature dependent modulus allowing change in dynamics as temperature changed during cooling, while the workpiece metal was assumed to be an elastic material.

SPH has been adapted to many other engineering and multi-physics applications [151, 152], including microfluidics and micro drop dynamics [153, 154], detonations and explosions [155], fracture of brittle solids [156], blood flow [157], gastric flow [158], traffic flow [159] and underwater explosions [160, 161]. A broad range of recent industrial applications, including casting are given in [96].

All these applications also can be modelled via Eulerian CFD. The major distinction between SPH and CFD is that SPH is mesh-free and Lagrangian, with the moving interpolation

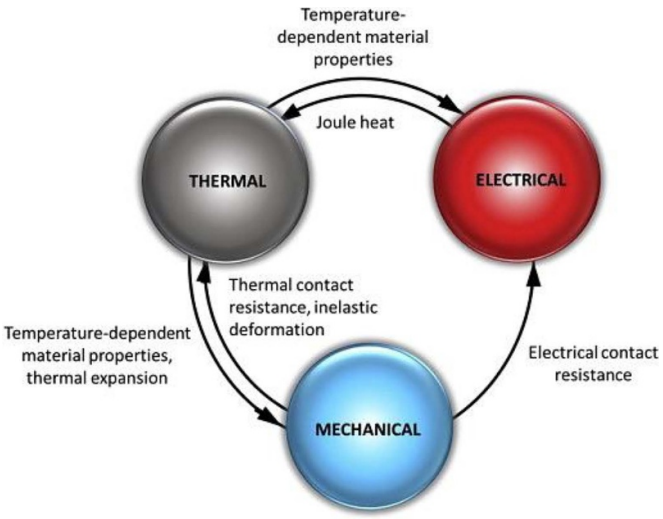


Figure 5. Schematic diagram demonstrating the thermal–mechanical–electrical coupling in FEA. Reprinted from [84], Copyright (2015), with permission from Elsevier.

points used in the spatial discretization carrying material mass and physical properties and usually referred to as particles. Accordingly, the method is commonly described as particle-based even though there are no physical particles involved and the equations being solved are derived from the same continuum governing equations that are solved in CFD. Unlike the Eulerian CFD method, the resolution of the simulation does not depend on the volume of the mesh elements; instead, it depends on the volumes of the virtual particles that make up the fluid and solids.

The absence of a mesh particularly suits SPH to simulating the free surface of highly deformable materials, such as molten metal, even when violent fluid flow leading to splashing and fragmentation occurs. SPH can model flow, heat transfer, deformation and material separation of high-strain-rate metallic materials (see [162]) in both liquid and solid phases, as used for WAAM fabrication.

4.9. FEA modelling of WAAM

FEA refers to the use of FEM to solve PDEs describing the governing physics of a system obtained as a function of space and time, such as those presented in section 4.3. It can predict the response of a solid structure (e.g. part or assembly) with complex geometry to complicated loading subject to sophisticated material laws (e.g. plastic deformation, creep, etc) driven by transient thermal loads. Although first used to solve structural mechanics problems in the mechanical domain, it has since been extended to other physics, including heat transfer, electromagnetism, sound, and vibration. Modern commercial FEM codes can support multiple physics and interactions between two or more types of physics, allowing more realistic simulations. For example, MSC Marc and Abaqus can simulate the thermal expansion of an electrical conductor due to Joule heating using fully coupled thermal–mechanical–electrical simulations (see figure 5).

FEM can also be used as a discretization strategy to model fluid flow (e.g. in commercial codes such as ADINA). However, the associated code cannot be concurrently used to simulate solid behaviour since the governing equations are entirely different.

In FEM, approximated 2D geometries (e.g. axisymmetric) or 3D geometries are divided into adjoining discrete (or finite) 2D or 3D elements, respectively. The PDEs are then approximated at nodes, typically (but not exclusively) placed at these elements' vertices. In addition to geometries, the analyst provides material properties, loads (e.g. heat flux, forces, electrical potential) that act on the system and boundary conditions at the external limits of the model. The FEM calculation then assembles the simultaneous equations and solves for the system's response to these loads. The output is in the form of a 'field variable' such as temperature, displacement or electrical current. In a coupled thermal–mechanical analysis, for example, both temperature and displacement are calculated. The next step is to calculate theoretical quantities by combining the nodal solutions with localized material properties. For instance, in the mechanical analysis, calculated displacements at the nodes of an element are converted to a stress profile within that element based on the (temperature-dependent) material properties of the element at that given instant based on interpolation functions, also known as shape functions. These describe how field variables such as displacement, temperature, and current vary within an element. Interpolations of field variables within elements are used to obtain secondary/theoretical quantities, e.g. the stress distribution within an element—from displacements converted to strains and using the latter's relationship with stresses via the Elastic Modulus.

In advanced FEA, 'nonlinear effects' that consider complex behaviour are included. Examples of nonlinear effects are contact between bodies, temperature-dependent material properties, nonlinear material behaviour (e.g. plastic, temperature-dependent), large changes in the shapes of bodies as the solution progresses, etc. FEA is a popular engineering tool as its discretization scheme supports the irregular geometries found in most real engineering situations. There are two methods of time integration used in FEA: implicit and explicit. In the implicit version, which is unconditionally stable, a nonlinear set of equations are solved to determine the state of a system at the current time step as well as a future time step. Although their stability allows for larger time steps, the nonlinearity makes it computationally intensive. By contrast, explicit methods, which are conditionally stable, compute the state of the system at the future time step directly through extrapolation from information known about the system at the current step. Implicit FEA ignores momentum effects, while explicit FEA accounts for forces due to acceleration (e.g. in a vehicle crash). In WAAM modelling, as in the simulation of many other manufacturing processes where the system remains at mechanical equilibrium, implicit FEA is used due to an absence of inertia effects.

When carrying out FEA, the analyst must first decide the nature of the analysis: thermal or thermal–mechanical. (While it is possible to consider a thermal–mechanical–electrical solution to include the resistive heating from current flow in the

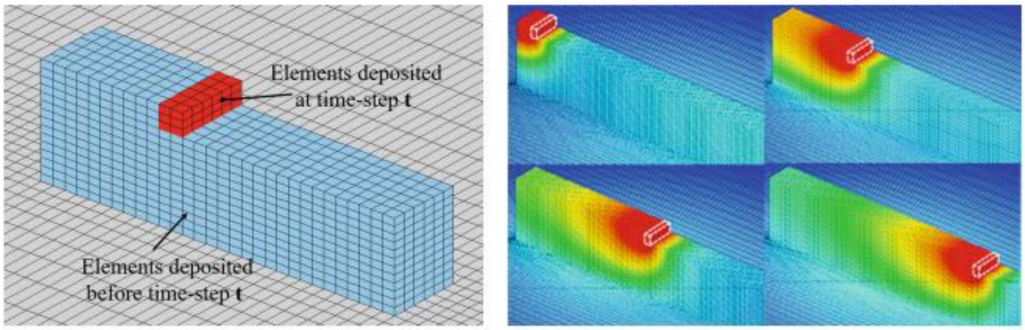


Figure 6. Finite elements (on the left) and temperature evolution in a thermal-only simulation. Reproduced from [129]. CC BY 4.0. The grey elements on the figure on the left represent the substrate. Note that the arc moves from the top left towards the bottom right.

metal, this is generally not included). During the analysis, the modeller defines the geometry of the substrate and the individual layers that are additively deposited. Note that, unlike in CFD, where the calculation predicts the shape of the deposit, the final shape must be provided to the FEA code. This is because fluid behaviour (e.g. effects due to surface tension or viscosity) is not accounted for. Once defined, these geometries are meshed appropriately, creating finite elements. This is followed by the assignment of initial temperatures, loads, boundary conditions and material properties. In the example shown in figure 6, from a thermal simulation by Chergui *et al* [129], the initial droplet temperature was calculated offline based on an assumption regarding the amount of heat available to melt the wire. The analyst assumed 50% of the arc heat Q_T was used to convert the wire into the droplet form, i.e. $Q_S/Q_D = 1$. (A 50/50 split of arc heat between the melting of filler material and heating of the substrate is commonly used—see, e.g. Montevecchi *et al* [163] and Dupont and Marder [164]). The droplet temperature would be above the liquidus temperature of the wire material; FEA does consider the latent heat of materials and allows it to be released over a range, i.e. between liquidus and solidus temperatures for alloys. The substrate's initial temperature (at time $t = 0$ s) is also provided, commonly room temperature. In the current example, the remaining 50% of the arc heat (Q_S) was input into the substrate using a Goldak double ellipsoidal volumetric model as a function of the arc position.

The next consideration is the strategy to be used in the simulation of the deposition process. The question is about how the deposit elements are to be treated up to the point where they become active, i.e. when the material represented by them gets deposited. There are at least two methods available for this: the inactive element method and the quiet element method. In the first, at the start of the simulation (i.e. $t = 0$), the user deactivates the deposit elements and then activates them progressively at different time steps based on the position of the arc (as illustrated on figure 6). The second approach involves keeping the deposit elements quiet—i.e. they are not deactivated but are initially given thermal and mechanical properties that are an insignificant fraction of their actual properties. As a result, they do not meaningfully take part in the simulation until their full properties are reinstated when the arc reaches their location. The inactive and quiet element methods have been reviewed comprehensively by Michaleris [165]. Finally, a suitable solver is chosen, and the solution is started. The FEA calculation solves for the temperature history of the simulation domain (right-hand side of figure 6).

5. Current status of WAAM modelling

CFD and FEA are the two numerical approaches currently most widely used in arc welding and WAAM modelling, while SPH has not yet been used to simulate WAAM. Due to the strengths and limitations of each approach, the scope of the simulation analysis and computational domain is largely predetermined, with the focus on heat and mass transfer in the melt-pool region and, in some cases, the arc for CFD and thermo-structural analysis for FEA.

CFD and SPH are effective in simulating both the flow of molten metal and its solidification during the WAAM process, making them valuable numerical tools for optimizing the process to prevent issues related to inertia, keyhole formation or porosity. Conversely, FEA can forecast temperature distribution, residual stresses and thermal distortion, making it beneficial for exploring heat-related problems. Certainly, various factors are linked to defects caused by inertia or heat—modelling each defect necessitates boundary condition strategies integrated with appropriate physics.

Tables 2 and 3 list summaries of selected recent articles presenting WAAM modelling using CFD and FEA, respectively. Since WAAM is based on arc welding, many aspects (including the suitability of specific materials for the feed wire and substrate, and numerical modelling approaches) of WAAM research articles overlap with previous publications focused on arc welding.

5.1. CFD approaches

CFD modelling of WAAM has built on a long history of arc welding modelling. Most CFD arc welding models have not included the arc plasma in the computational domain but nonetheless provided valuable insights into factors including penetration depth in complex geometries [166, 167], humping phenomena [168] and defect formation [169]. Early MHD

Table 2. Summary of CFD modelling of the WAAM process.

References	CFD Code	Simulation scope	Model validation	Simulation conditions	Study case and remarks
Cadiou *et al* [177]	COMSOL, coupled with the level-set method	CMT-WAAM process modelling, focusing on the melt pool and droplet transfer regions.	Comparison with measured melt-pool length, and width and height of deposited layers	Constant current of 25 A, transfer frequency of 15 Hz, stainless steel	Electromagnetic, heat transfer and fluid flow in the plasma and the melt pool were simulated. The forward and backward motion of the wire was considered. The plasma properties were very approximate. Good agreement was found between measured and predicted melt pool length.
Ikram and Chung [178]	COMSOL employing MHD and phase-field method	2D simulation of the metal droplet detachment including the arc plasma.	Transient droplet profiles were compared with simulation data.	Current pulse frequency range of 50–225 Hz with peak current of 380 A, mild steel	The detachment pattern of droplets for different pulse frequencies was presented. Droplet deposition was not considered.
Ke *et al* [179]	ANSYS FLUENT, incorporating VOF method	3D multi-layer simulation of pulsed WAAM of NiTi shape memory alloys	Simulated melt-pool profiles compared with physical bead profiles for one to five layers	Transient linear multi-layer deposition, with analytic heat source model; constant current of 100 A compared to 20 kHz pulsed current	Coupled experiment-simulation approach to study multi-layer deposition of NiTi shape memory alloys. Microstructure properties were observed by experiment, and the temperature and flow in the melt pool were predicted by CFD.
Ou *et al* [119]	In-house FORTRAN code	Multi-track depositions of steel H13 and titanium alloy Ti-6Al-4V were compared via a transient finite difference control volume method.	Various bead shapes in cross and transverse sections were compared with the numerical data.	Constant current of 150 or 200 A. A analytic heat source model was used, and the influence of droplets was averaged over time.	Ti-6Al-4V was shown to have a larger fusion region than that of H13 steel, as density of H13 steel is much larger than the density of Ti-6Al-4V. An adaptive mesh algorithm provided higher computational efficiency.

(Continued.)

Table 2. (Continued.)

References	CFD Code	Simulation scope	Model validation	Simulation conditions	Study case and remarks
Zhao *et al* [180]	ANSYS FLUENT, incorporating VOF method	3D GMAW-WAAM multi-layer deposition simulation employing dynamic mesh technique to model the droplet detachment and to expand the air subdomain.	Three linear builds were produced to compare with numerical models: single-pass single layer, single-pass two layers and single-pass five layers.	Constant current of 135 A, aluminium alloy; Goldak's double ellipsoid heat source model was used.	Metal deposit morphology and droplet transfer were modelled and showed good agreement with experiment. The formation and detachment mechanism of a globular droplet at the tip of the moving wire were coupled to the melt pool.
Zhao *et al* [181]	ANSYS FLUENT, incorporating user-defined scalars and user-defined functions for defining source terms	3D GMAW-WAAM process focusing on single-pass single-layer deposition with comprehensive arc physics. Metal vapour was also considered.	Droplet transfer processes at different times recorded with a high-speed camera were compared with the simulation data. The cross-section profile of a single layer was also compared.	Al-5%Mg was used for a feed material with temperature-dependent thermal conductivity and viscosity. A constant current of 135 A was used.	The model demonstrated that the consideration of Al vapour significantly reduced the arc temperature, resulting in a low-temperature region at the centre of the arc. Most of the Al vapour originated from the molten wire tip; the Al vapour generation from the surface of the melt pool was negligible.
Li *et al* [182]	FLOW-3D, incorporating VOF method	3D GMAW-WAAM process using low alloy steel, simulating up to 5-layer deposition. Single-pass and square cladding passes were simulated.	The model was validated by comparing 3 different welding positions in single-pass single-layer morphologies of down-hand welding, horizontal welding and vertical welding.	3 different welding current values between 140–220 A were used with different wire feed rates (3.4–6.5 m min^{-1}).	Molten metal droplet deposition, fluid flow in the melt pool and solidification were considered but the arc physics was not considered. The model demonstrated that single-direction welding caused excessive heat accumulation at the arc-quenching position, resulting in an inconsistent height.

(i.e. CFD combined with electromagnetic equations) arc models did not include the electrodes in the computational domain (e.g. [170, 171]). The subsequent models evolved to include the melt pool and electrode in GTAW (e.g. [138]), and droplet transfer [172, 173], metal vapour [135, 174] and mixing of droplets in the melt pool [22] in GMAW. Including the arc plasma in the computational domain provides more detailed insights into the interactions of the arc with the electrodes and removes the need to perform experiments to determine the free parameters in the expressions for the heat flux, arc pressure and electromagnetic force. However, the complexity of the model increases significantly, increasing computational time and memory requirements, although appropriate approximations can reduce this problem [175]. While a wide range of welding parameters, wire orientations and alloys can be treated [21, 176], modelling of WAAM also requires the capacity to model the deposition of several layers, which is a challenging extension.

Some representative examples of CFD simulations on WAAM are summarized in table 2. Ikram and Chung [178] investigated the formation and detachment of the metal droplets during WAAM and its profile using a 2D MHD simulation. The computational domain included the wire, droplets and the arc plasma, but not the substrate. The simulation was conducted by using the commercial CFD code COMSOL Multiphysics 5.6, in which the governing equations for the fluid regimes were solved using the FEM. The phase field method was adopted to allow the evolution of the droplet shape to be calculated. The primary focus was on the effect of the pulsed current waveform on the droplet detachment, with pulse frequencies varied from 50 to 225 Hz. Four different metal transfer modes were identified, including the desirable one-drop-per-pulse mode. The Lorentz force and surface tension were identified as contributing towards droplet detachment, which is in accordance with studies of GMAW [183]. The neglect of metal vapour in the arc leads to unrealistically high plasma temperatures, which will affect the reliability of the droplet predictions.

Cadiou *et al* [177] simulated the initial layer of metal droplet deposition and the temperature field in the melt pool and heat-affected zone. Like Ikram and Chung [178], they used the COMSOL Multiphysics software and included the arc plasma in the computational domain. However, the simulation was 3D, and the level-set method was used to track the free surface of the molten metal. The authors made several notable assumptions to reduce the complexity of computations, which include neglecting the Marangoni convection in the melt pool and the influence of metal vapour in the arc, and assuming the current was a constant 25 A, which is an unrealistically low value. The density, thermal conductivity and viscosity of the plasma were assumed to be independent of temperature when, in reality, they vary by orders of magnitude. The low current and inaccurate plasma properties led to temperatures that were unrealistically low for a pure argon plasma; however, this partially compensated for the neglect of metal vapour, whose presence cools the plasma. The predicted melt-pool length was in accordance with measurements, but the build width was overestimated. Only a single set of parameters was considered in the comparison. The authors noted that the computational time (15 d to simulate 6 s) prohibited the simulation of the production of a part.

Zhao *et al* [180] conducted a 3D multi-layer simulation of GMAW-based WAAM. The commercial CFD software ANSYS FLUENT, a finite-volume-based code, was used. The droplet detachment and transfer to the melt pool were modelled using the volume-of-fluid (VOF) method. The arc plasma was not modelled but was accounted for by a double ellipsoidal heat flux and analytic expressions for the arc pressure and electromagnetic forces similar to those given in equations (31) and (33)–(35), respectively. Results were presented for the deposition of a single layer, two layers and five layers, all along a linear path. Reasonable agreement was demonstrated between the measured and predicted shapes of the build cross-sections. The Marangoni force was found to drive outward flow near the top surface of the melt pool. The height of the second layer was approximately 10% lower than the height of the first layer because the deposited layer provides less resistance to the droplet impingement than the substrate. This effect can be counteracted by allowing the layer to cool before deposition of a subsequent layer. The paper demonstrates the value of a CFD simulation that considers the droplet transfer, the melt pool and the substrate, particularly when the deposition of several layers is simulated.

Ke *et al* [179] used a similar approach to Zhao *et al* [180] (ANSYS FLUENT and VOF, analytic expressions for the heat flux, arc pressure and electromagnetic forces) to model the deposition of up to five layers. Unlike Zhao *et al*, the authors did not model droplet growth at the tip of the wire. Also, to save computational time, they represented previous layers by a quarter-elliptical approximate form of the previous layer. A constant current of 100 A was compared to a 20 kHz pulsed current with the same average current. Experiments revealed that the ultrahigh-frequency current increased grain refinement, and the model was used to explain this in terms of the induced fluctuations in the flow velocities in the melt pool. Ke *et al* also found that the molten pool grew in length as layers were deposited due to the longer heat transfer path to the substrate and, as did Zhao *et al*, that the first layer was the highest.

Ou *et al* [119] investigated the multi-layer deposition of Ti-6Al-4V and H13 tool steel using a using a self-written Fortran code that modelled flow and heat transfer in the melt pool. Analytic expressions for the arc heat flux, arc pressure and droplet pressure were adopted, but the electromagnetic force was ignored. The fusion zone was larger for Ti-6Al-4V, which was attributed to its lower density, but the mushy zone was smaller because of the smaller difference between the liquidus and solidus temperatures. The authors demonstrated that neglecting fluid flow led to a large underestimate in the build width for H13, for which the Peclet number (the ratio of convective to conductive heat transfer) was larger than for Ti-6Al-4V. Arc pressure was found to be more important than droplet pressure and the Marangoni force in determining the build height. The influence of hatch spacing (the distance between adjacent layers) was investigated, with a smaller spacing increasing melt

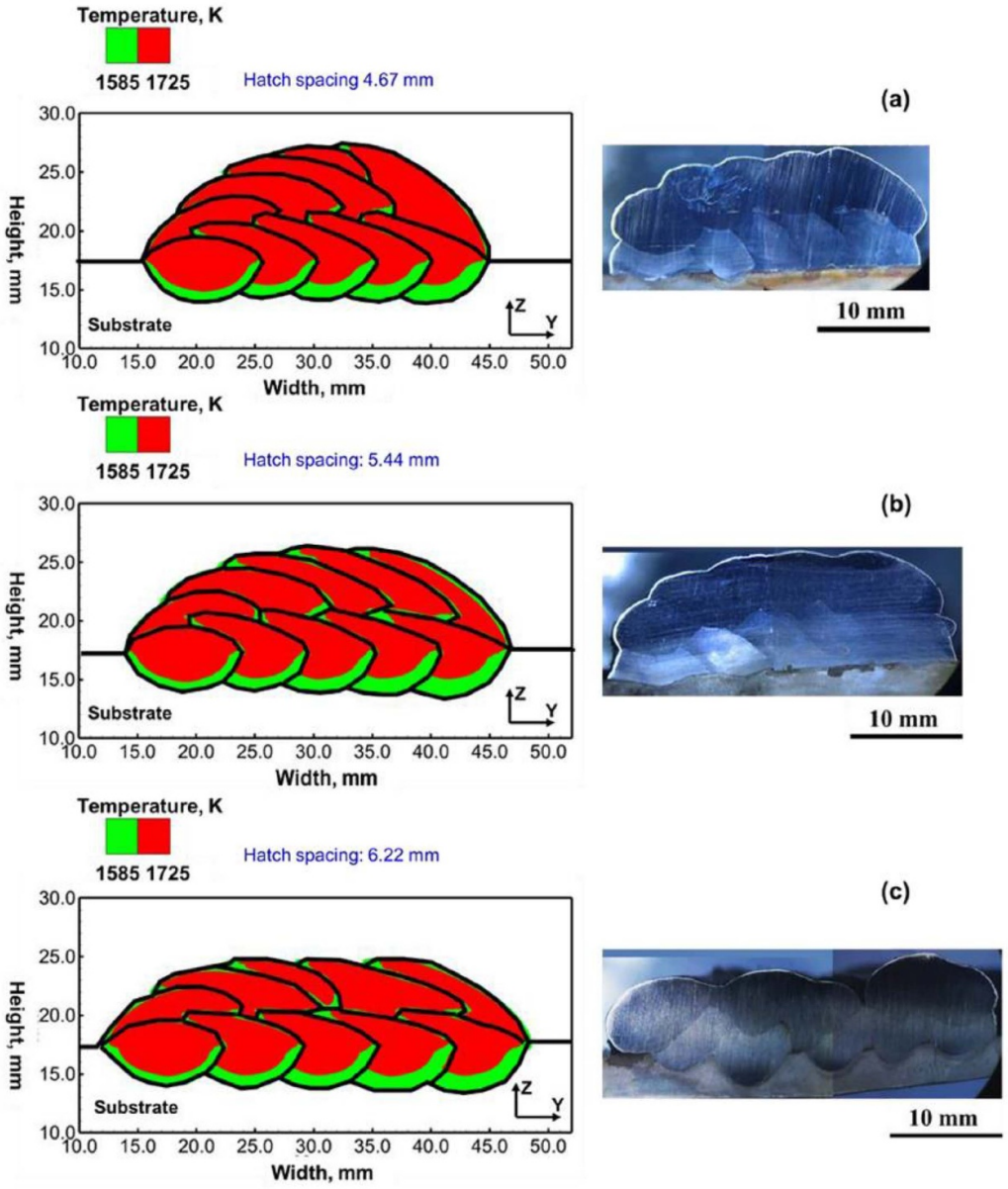


Figure 7. Comparisons between predicted and measured transverse build section for two-layer, five-hatch deposit of H13 steel, for hatch spacings of (a) 4.67 mm, (b) 5.44 mm and (c) 6.22 mm. The colour scale indicates the solidus and liquid temperatures, 1585 K and 1725 K, respectively. Reprinted from [119], Copyright (2021), with permission from Elsevier.

pool size if the previously deposited tracks were still hot, as shown in figure 7. The results indicated the value of modelling in predicting and explaining results using different metals and optimizing the toolpath planning in WAAM.

There are numerous examples of CFD modelling of arc welding in which the wire, droplets, arc (including metal vapour effects in some cases), melt pool and substrate are all included in the computational domain. However, the examples above indicate that a 3D simulation of a multi-layer WAAM build that includes all the elements is still computationally prohibitive. Nevertheless, appropriate approximations and assumptions can be introduced that allow CFD models to give reasonable agreement with measurements, provide valuable insights, and have significant predictive power.

5.2. SPH approaches

SPH model of WAAM is still not yet available in the public domain. Here, we discuss selected examples from the literature to illustrate how SPH has been applied to simulate arc welding, to address the potential of SPH modelling for WAAM process.

SPH simulation of arc welding was first implemented by Das and Cleary [20, 99, 184, 185] to model plastic deformation and thermal responses in the heat affected zone (HAZ). The plastic strain of the melt pool and the temperature field on HAZ were calculated to predict the deformation of the welded area. For simplicity, the authors assumed that the computational domain consisting of the deposited material and

the entire substrate is solid. Shigeta *et al* [186] instead considered the flow in the weld pool, considering all the most important driving forces: the Marangoni effect, the Lorentz force, buoyancy and the drag force acting on the surface of the melt pool. The simulation predicted the temperature distribution in the melt pool, and highlighted the effect of the sulfur content, which affects the Marangoni flow by altering the temperature dependence of surface tension flow.

Komen *et al* [187] applied a 3D SPH model to a high-current GMAW process known as buried arc welding, in which metal transfer from the wire occurs in a space under the surface of the weld pool. They modelled two different deposition modes: drop transfer mode, in which large droplets are detached axially, and rotating transfer model, in which the liquid wire tip rotates, leading to the detachment of smaller droplets. The incompressibility of the molten metal was treated using the density homogenization method of Shigeta *et al* [188]. The molten metal droplet was modelled using 64 particles, equivalent to 0.003% of the total particles in the computational domain. While the heat source distribution is expected to depend on the metal transfer mode, the authors chose to use the same simplified heat source expression for both transfer modes with the aim of evaluating the influence of the differences in the arc pressure and droplet direction. It was shown that both the arc pressure and the droplet impact influenced the formation and collapse of the buried cavity. The study demonstrated the ability of SPH to simulate the complex flow phenomena, including the interactions between the droplets and weld pool.

Most recently, Komen *et al* [189] demonstrated a combined approach by investigating heat transfer phenomena occurring during submerged arc welding (SAW). In this process, the wire electrode tip and the weld pool are submerged under a flux powder, which melts to form a slag that floats on the weld pool. The process is challenging to model because of the presence of particles, solid metal, and two liquid materials, and the need to consider the heat transfer and transitions between the different material phases. The formation of the weld pool and the slag were simulated using SPH coupled with DEM. The heat flux from the arc, which was represented as a GMAW arc with a flat workpiece, was predicted via an FVM-based CFD model of the arc plasma and applied in the SPH model as a boundary condition. The use of a GMAW heat source, which is a significant simplification, was justified by comparing with experimental data indicating that the thermal efficiencies of GMAW and SAW arcs are similar [164]. The multi-simulation model is a good example of a Eulerian–Lagrangian combined approach, where two distinct Lagrangian methods were coupled to simulate the heat transfer and transitions between the solid and liquid phases, while the Eulerian method was used to calculate the heat source distribution from the arc. The complexity of the calculation is demonstrated by the results given in figure 8, which shows heat flowing from the metal to the flux in the blue region, surrounded by a larger red region in which the flow of heat is in the other direction. Good agreement was found between the predicted and measured weld-bead profiles. The authors determined that the combined model provided useful results despite the inherent simplifications.

The application of SPH to the modelling of powder-based AM also demonstrated features relevant to WAAM. Cummins *et al* presented a DEM-SPH model simulating the microstructure evolution in the laser powder-bed fusion process using Ti-6Al-4V [23]. In their work, DEM and SPH are run in chronological order, and there are no concurrent interactions between the methods. A phenomenological approach to predict the solid titanium phase was incorporated. The authors demonstrated interparticle reactions such as collisions between non-spherical grains during the coating process via a spring-dashpot contact model of DEM, while SPH was used to simulate the heat transfer from the laser, powder melting and flow, solidification of the molten metal, and multi-phase interactions between the molten and solid powders. DEM is not required in WAAM modelling since the feedstock is a metal wire, but heat transfer from the arc, melting and solidification processes, and multi-phase interactions between the molten metal droplet and the solid substrate can be realised by SPH in a similar manner.

Qiu *et al* also used SPH to simulate the surface tension and wetting effect occurring in the melt pool in laser powder-bed fusion. The metal powders were treated as agglomeration of SPH particles, but no description of agglomeration factors was provided [190]. The authors highlighted that the Marangoni forces and recoil pressure are critical factors determining the melt pool morphology, and the cubic-droplet-to-sphere transformation and droplet-wall interaction were used to verify the surface tension and wetting effect. These are also crucial aspects in the SPH model of WAAM, determining the melt pool profile and multi-phase interaction between the molten metal droplet and the substrate or pre-deposited layers.

These examples demonstrate that SPH can model the dynamics of droplet deposition and melt pool flow, including in combination with other simulation methods. It can also predict the temperatures of both fluid and solid phases concurrently. The Lagrangian formulation allows complex processes to be simulated relatively easily. SPH is a suitable method to model the thermal and fluid flow phenomena in WAAM, and that it can be combined with other modelling approaches.

5.3. FEA approaches

FEA provides a means for predicting the solidification behaviour during the WAAM process and the subsequent cooling of the part to room temperature, resulting in residual stress development and distortion during the build and after being separated from the substrate (or build plate). A further strength of the method is its ability to support complex shapes, making it a method of choice for manufacturing process simulations of real-world production parts. Additionally, because of the inherent simplifications, such as neglect of flow, FEA simulations are typically faster than CFD and SPH. Thus, a good number of investigations available in the open literature have used FEM to simulate a simplified representation of the WAAM process, although some were purely thermal versions that ignored the mechanical aspects. Amongst the thermal–mechanical simulations, some used a fully coupled strategy

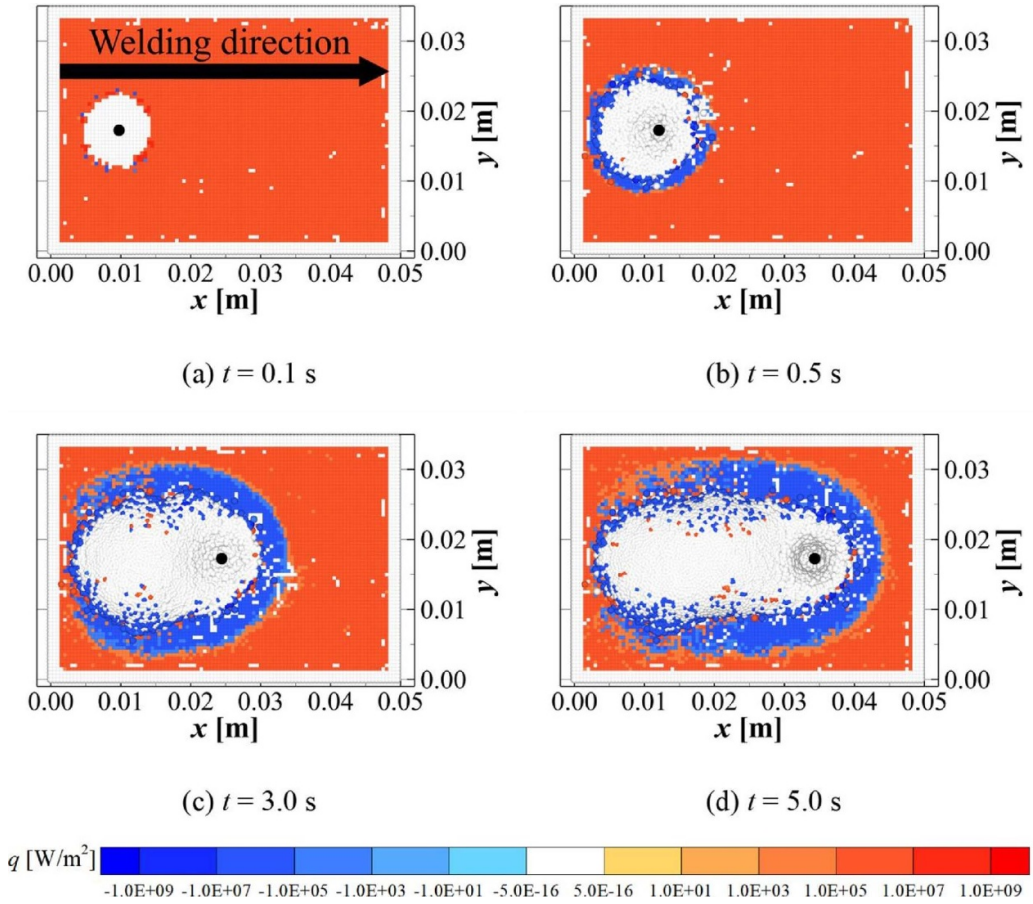


Figure 8. The heat flux distribution from the flux to the metal surface in the SPH-DEM model of submerged arc welding. Red denotes heating of the metal, blue cooling of the metal, and the black circle shows the centre of the arc heat source; the white region corresponds to slag (molten flux). Reprinted from [189], Copyright (2021), with permission from Elsevier.

for linking the thermal and mechanical physics, while others used the less accurate sequential coupling. In the latter, the entire temperature histories at nodes were first obtained from a thermal analysis and input into the mechanical analysis. Thus, any changes to the temperatures from mechanical effects (e.g. plastic heat generation) were not considered in sequential coupling. Some articles did not mention if a fully coupled or sequentially coupled strategy was used.

While some studies used the predictive power of FEM to develop novel insights into the various aspects of the WAAM process, a small fraction focused on other matters, such as improving the computational efficiency of the solution and enhancing the quality of inputs, furthermore, a limited number of models were not experimentally validated.

A representative sample from the more recent WAAM-related FEM works are summarized in table 3.

We now briefly review selected studies to provide the reader with a flavour of the breadth and depth of modelling conducted in this area. Firstly, we focus on studies that used the predictive power of FEM to generate novel insights into the WAAM process and move on to those that aimed to improve the FEM solution process itself, e.g. by increasing the efficiency of computations.

FEA is a powerful tool for determining the optimum heat management strategies that can minimize residual stresses and, therefore, distortion of the part after being removed from the substrate. Wu *et al* [198] developed a thermal–mechanical model to study the effect of three deposit paths (long, short, spiral) on residual stresses and distortion in two materials: Ti-6Al-4V and Inconel 718. They found that depositing short path lengths can minimize residual stresses and distortion among the three patterns investigated for both alloys. This was because heat was distributed in the entire set-up (as opposed to being concentrated in a location), reducing temperature gradients that give rise to residual stresses. In a similar but purely computational thermal–mechanical study, Amal *et al* [195] investigated the effect of the deposit path strategy employed (raster, two-section raster, zigzag, spiral, zigzag + spiral) on residual stresses and distortion. They also concluded that paths that encouraged more uniform cooling resulted in lower residual stresses.

FEA has been used in the design of manufacturing setups for the WAAM process. Veugelers [200] used a validated fully coupled thermal–mechanical model to simulate the WAAM process to design a clamping system for the substrate that would minimize part distortion. He succeeded in proposing

Table 3. Summary of numerical modelling cases for analysing WAAM process using FEM.

References	FEM code	Simulation scope	Model validation	Simulation conditions	Study case and remarks
Ahmad *et al* [128]	MSC Marc	Distortion and residual stress analysis	N/A	Rectangular bead shape, five layers and three strings.	Goldak's double ellipsoid and rectangular heat source models were adopted and showed similar distortion behaviour. The residual stresses were strongly affected by the transient temperature distribution.
Casuso *et al* [191]	FEA	Deformation prediction of GMAW-WAAM manufactured thin-walled parts.	Physical geometry and CAD model are compared for geometric verification using the laser tracker.	Cold-arc transfer mode was adopted for the first ten layers, then pulsed-GMAW was adopted for latter ten layers.	Experimentally measured deformation data (obtained via the laser tracker measurement) compared with the predicted deformation data. The temperature distribution showed deviation of less than 5%.
Ding *et al* [101]	Ansys	Thermal simulation of the CMT-WAAM process: Temperature distribution and thermal cycle prediction along the substrate	Maximum temperatures taken from the build and temperature histories from the substrate were compared with predictions that assumed a cuboid shaped heat source	Ten layers of the build. Wire electrode = low alloy steel and substrate = Q235B steel.	Three different heat source shapes (double ellipsoidal, hemispherical with a Gaussian profile and a cuboid) were used. The authors claimed the highly approximated cuboid shape gave the best computational efficiency (91% saving over double ellipsoidal using a moderate mesh) while predicting temperature histories on the substrate with acceptable accuracy.
Zhou *et al* [192]	FEA, ML	Residual stress prediction using coupled thermo-mechanical model and ML.	Residual stresses were measured to validate FEA model.	1.5 kW of power applied under moving speed of 10 mm s^{-1}.	A 3-level data-driven method proposed to model residual stresses of WAAM products.

(Continued.)

Table 3. (Continued.)

References	FEM code	Simulation scope	Model validation	Simulation conditions	Study case and remarks
Chergui *et al* [129]	Not mentioned. Presumably an in-house code.	Thermal simulation of the CMT-WAAM process.	Temperature histories from the model were compared with those of three thermocouples on the surface of the substrate.	Two eight-layer thin walled 5356 Al parts were built on 5083 Al substrate.	Goldak's double-ellipsoid and rectangular heat source models were used. Reasonable agreement was obtained between predicted and experimental temperature histories.
Heinrich *et al* [193]	Ansys	Thermal simulation of WAAM using gas metal arc welding (GMAW). Use of forced cooling on substrate.	Predictions of temperature histories were validated with experimental data from thermocouples on the substrate	Wire = SUPERARCL-50 (copper coated mild steel) and substrate = A36 hot rolled steel. Layer height = 2.1–2.5 mm.	The heat source details were not given. It is likely the deposits were introduced at their melting temperature of 1500 °C. The model results showed that the introduction of a forced cooling arrangement in the substrate can accelerate the cooling of the WAAM part.
Yang *et al* [194]	Abaqus	Sequentially coupled thermal–mechanical model with semi-analytical thermal model to improve computational efficiency. CMT-WAAM with a modified GMAW variant.	Thermal and mechanical predictions were validated with experimental data and modelling results available in the literature	A thin wall structure of 500 mm length and 4 mm thickness. A total of 4 layers. Wire = mild steel and substrate = S355JR-AR mild steel.	Goldak's heat source was used. The semi-analytical thermal approach reduced solution times from 8 h to 3 h 15 min using coarser meshes. However, the strategy is suitable only for materials whose residual stresses and distortions due to WAAM are not sensitive to the phase transitions and solid phase transformations. Also, the method is highly sensitive to assumptions relating to boundary conditions.

(Continued.)

Table 3. (Continued.)

References	FEM code	Simulation scope	Model validation	Simulation conditions	Study case and remarks
Amal *et al* [195]	MSC simufact welding	Thermal–mechanical model focused on studying the influence of deposit path on residual stresses and distortion.	N/A	A cuboid shape with a short height ($175 \times 141 \times 10\,\text{mm}^3$) was constructed. Wire and substrate = 16MnCr5-SPM-SW alloy steel	Glodak's double ellipsoidal heat source model was used. The study concluded that deposit paths that encouraged more uniform cooling resulted in lower residual stresses.
Gornyakov *et al* [196]	Abaqus	A hybrid steady state/transient model was used to reduce solution times. Simulated WAAM and high pressure rolling. Sequential coupling of physics by design.	Temperature histories were validated with information available in the literature.	A 20-layer wall was simulated. Wire and substrate = S355 mild steel.	Goldak's double ellipsoidal heat source model was used. The authors achieved an 83% improvement in computational time compared with transient simulations of the WAAM process (and 96% for high pressure rolling) by using a steady state model to estimate temperatures.
Ding *et al* [5]	Abaqus	A hybrid steady state/transient model was used to save on solution times. The aim was to predict distortion and residual stresses. Sequential coupling of physics by design.	Temperature histories as well as residual stresses and distortion were validated using experimental data.	A 4-layer wall was simulated. Wire = low carbon steel and substrate = S355JR-AR structural steel	Goldak's double ellipsoidal heat source model was used. The authors claimed that their newly proposed FEM approach based on the steady-state thermal model showed a significant advantage of around 80% on the computational time.
Montevecchi *et al* [197]	LS-Dyna	An adaptive remeshing strategy was used in a thermal–mechanical simulation of WAAM. An algorithm was used to remesh the domain during the solution process.	Distortion predictions were validated with experiments.	A 5-layer wall was simulated. Wire = ER70S-6 low carbon steel and substrate = S235JR structural steel.	The scheme proposed in [137] was used. Using the remeshing strategy, the modellers completed the thermal–mechanical simulations in 30% of the time required to carry out simulations without remeshing.

(Continued.)

Table 3. (Continued.)

References	FEM code	Simulation scope	Model validation	Simulation conditions	Study case and remarks
Montevecchi *et al* [163]	LS-Dyna	Thermal–mechanical model of gas metal arc welding (GMAW) WAAM to test a newly proposed heat source model.	Displacements predicted by models were validated with experimental data.	A wall was simulated, but the number of layers was not mentioned. Wire = ER70S-6 low carbon steel and substrate = S235JR structural steel.	The authors proposed decoupling the arc heat into heat used for melting the filler material (50%) and heat directed on the substrate (50%). The authors concluded simulation results incorporating this assumption provided better agreements with experimental data than those obtained using the entire arc heat on the substrate. The Goldak model was used for the heat directed on the substrate.
Wu *et al* [198]	Abaqus	Thermal–mechanical model of the WAAM process.	Residual stresses and distortion were validated using experimental data.	A 5-layer cuboid shape was made. Wire and substrate were both Ti-6Al-4V for the first set and Inconel 718 for the next.	Goldak's double ellipsoidal heat source model was used. The study noted that, since residual stresses primarily depend on the temperature variation during cooling of the solids, the convective flow may be neglected in the thermal–mechanical model to make the calculations tractable. The study also showed that printing with short track lengths can minimize residual stresses and distortion among the three patterns investigated for both alloys.
Ding *et al* [199]	Abaqus	Sequentially coupled thermal–mechanical simulations of the WAAM process. Using a hybrid strategy involving steady state and transient models.	Temperature histories as well as residual stresses and distortion were validated using experimental data.	A 4-layer wall was simulated. Wire = low carbon steel and substrate = S355JR-AR structural steel	Goldak's double ellipsoidal heat source model was used. The authors claimed their hybrid thermal–mechanical simulations gave a 99% saving in computation time over transient simulations.

(Continued.)

Table 3. (Continued.)

References	FEM code	Simulation scope	Model validation	Simulation conditions	Study case and remarks
Veugelers [200]	MSC Marc for thermal–mechanical simulation of WAAM and Ansys for a static mechanical simulation for the design of clamping	A fully coupled thermal–mechanical model to design a clamping system for WAAM substrate to reduce part distortion.	The predicted distortion of the part was validated against experimental observations.	A 20-layer wall was simulated. Wire and substrate = Ti-6Al-4V	Gaussian heat source was used. The work resulted in a clamping system design being proposed to contain part distortion. The author stated that the design was of a modular nature in that it could be used for substrates of arbitrary dimensions.
Farias *et al* [24]	MSC simufact welding	Fully coupled thermal–mechanical simulations were conducted to provide data points for training a ML model.	Temperature histories were validated with experiments.	A 20-layer wall was simulated. Wire = AWS ER90S-B3and substrate = not mentioned	Goldak's double ellipsoidal heat source model was used. The model's predictions were used in an Artificial Neural Network (ANN) ML model to predict optimum interpass temperatures.
Gupta and Mulik [201]	Ansys thermo–mechanical module	Transient thermal–mechanical simulations were performed to model the thermal history along the rectangular welding path.	Temperature histories were validated with experiments.	A 4-layer rectangular wall was simulated. Wire and substrate were mild steel	Goldak's double ellipsoidal heat source model was applied to the rectangular-shaped deposition. The transient temperature changes at various points on the substrate are examined.

a modular design that could accommodate substrates of any dimension. Similarly, Heinrich *et al* [193] created a validated thermal FEM model to design a forced cooling arrangement for a steel substrate and to determine the optimum dwell time between layers. They found that the steel part (a hollow 122 mm square vertical wall with 12 mm thickness and 89 mm height) that was built using a GMAW arrangement cooled faster when the forced cooling was active. It was likely that the deposits were introduced at 1500 °C, the melting temperature of steel. However, the authors failed to mention if a heat flux was applied to the previously deposited layers.

While Goldak's double ellipsoidal heat source model was used in most cases to account for heat from the arc, Montevecchi *et al* [163] identified a shortcoming of the model. This is related to the fact that the model is agnostic to the power distribution between the filler and the substrate. To address this, they recommended decoupling the fraction of the power used for melting the filler material from that used to heat the substrate (as explained in detail in section 4.6.1). They recommended using the Goldak model for the heat delivered directly to the substrate and proposed using the balance of the heat for melting the filler material. This model was used to supply internal heat generation to the deposited elements to obtain the droplet temperatures (unlike in our explanation, where this calculation was carried out manually offline). The authors validated the distortion predictions of their thermal–mechanical model with and without the decoupling and concluded that the predictions that included the separated heat sources were more accurate. They recommended using equal amounts of the arc heat for melting the filler material and heating the substrate.

Some studies have sought ways to accelerate the FEM solution process. Despite FEA being generally faster than analyses based on CFD or SPH, the fact that a metres-scale WAAM part is built one small layer (of the millimetre scale) at a time makes its solution wall time typically run into hours or days depending on the size and complexity of the model.

This is primarily because numerous small elements need to be defined, requiring simultaneous equations to be solved at a very large number of nodes at each timestep. In addition, the timesteps need to be sufficiently small to capture the transient phenomena that occur during the build process. Thus, finding strategies to accelerate the solution process has been the aim of some investigations. One such work was that of Ding *et al* [101], who compared the computational efficiencies involving three heat source shapes—double ellipsoidal, hemispherical with a Gaussian profile and cuboid. They claimed that the highly approximate cuboid shape gave the greatest efficiency while providing good agreement with experimental temperature histories obtained from the substrate. However, unfortunately, they did not report the temperature histories obtained from the assumption of double ellipsoidal or hemispherical heat source shapes; thus, it was not possible to ascertain the relative accuracies attributable to the assumptions on shapes. In addition, the experimental temperature histories that were compared were from the substrate rather than from the build itself—thus diminishing the value of the comparisons. Ding *et al* did, however, compare the *maximum* temperatures recorded on the build by a thermal camera with their predictions using the cuboid heat source on the tenth layer at locations corresponding to the start, middle and end of the layer. The simulated maximum temperatures were, respectively, 89 °C, 166 °C and 74 °C higher than measured maximum temperatures. Thus, it is hard to draw concrete conclusions solely from this work on the 'computational accuracy vs computational efficiency' relationship attributable to the assumption on heat source shapes.

Another recent work that sought to study computational efficiency is that of Yang *et al* [194], who developed a thermal–mechanical model. Their model used a semi-analytical thermal approach designed to reduce solution wall time. An analytical thermal model accounted for the Goldak heat source, and a complementary temperature field accounted for the boundary conditions. Temperature evolution was then coupled to an elastoplastic mechanical boundary value problem, which computed thermal stresses and distortions. Their model solved in 3 h and 15 min using a coarse mesh, while a traditional thermal–mechanical simulation that required a finer mesh took 8 h. The accuracy of the analytical approach was acceptable. However, their approach has some major limitations. The authors state that, since the non-linearity during the thermal simulation is not considered, the values of thermal properties need to be carefully determined by matching the predictions with experiments. Additionally, their model is suitable only for materials whose residual stresses and distortions due to WAAM are not sensitive to phase transitions or solid phase transformations. It also suffered from the high sensitivity of its distortion predictions to assumptions involving boundary conditions.

Other strategies were also used as part of the attempts to shorten solution times. Some analysts used hybrid approaches where transient models that provided histories of state

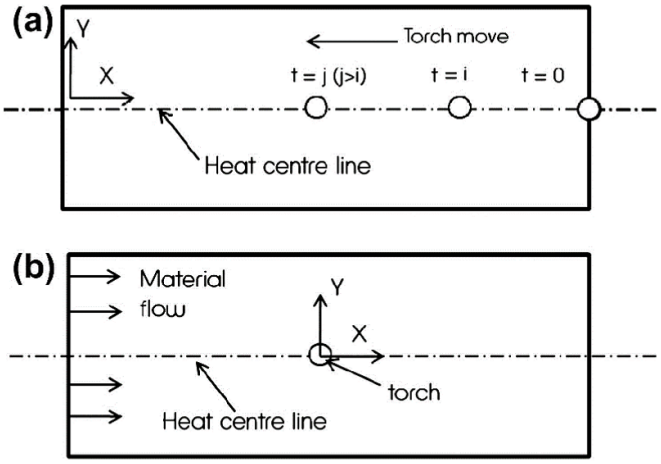


Figure 9. Model with different reference frames. (a) Lagrangian reference frame. (b) Eulerian reference frame. Reprinted from [5], Copyright (2011), with permission from Elsevier.

variables such as temperature and residual stresses using time-marching schemes were combined with the rapidly solving steady-state models. For example, hybrid models reported by Ding *et al* [5, 199] achieved savings of 80% and 99% in solution times over purely transient versions without significantly compromising the accuracy of temperature and distortion predictions. Their plan involved first calculating steady-state temperature profiles using the Eulerian thermal method. In this scheme, a Eulerian reference frame was attached to the moving arc, and material flows through the mesh, as shown in figure 9. Therefore, the problem can be solved for a single time step—resulting in a significant saving in computational time. Further savings were obtained using coarser meshes. The steady-state temperature field was then used as input to a 3D mechanical model for residual stress and distortion analysis. Since the steady-state model did not have a temperature history, the analyst had to obtain a virtual history by post-processing the temperature results along the path of the arc (x-direction figure 9) to simulate the passage of time based on arc speed. Similarly, Goryakev *et al* [196] used a hybrid strategy to accelerate thermal–mechanical solutions of WAAM simulations. They used a small part of a part geometry to generate a steady-state temperature profile, which was then transferred to a mechanical model to arrive at the mechanical solution for displacements. These were then mapped onto the entire part using a strategy shown in figure 10. The approach required selecting appropriate boundary conditions in the solution for the smaller part. The researchers found that the model took only 17% of the time needed for the full transient version.

While hybrid strategies, as demonstrated above, are useful alternatives to consider for obtaining quicker results, they are far from elegant. They require significant additional manual work by the analyst for simulation set-up, savvy treatment of boundary conditions, mapping between the temperature

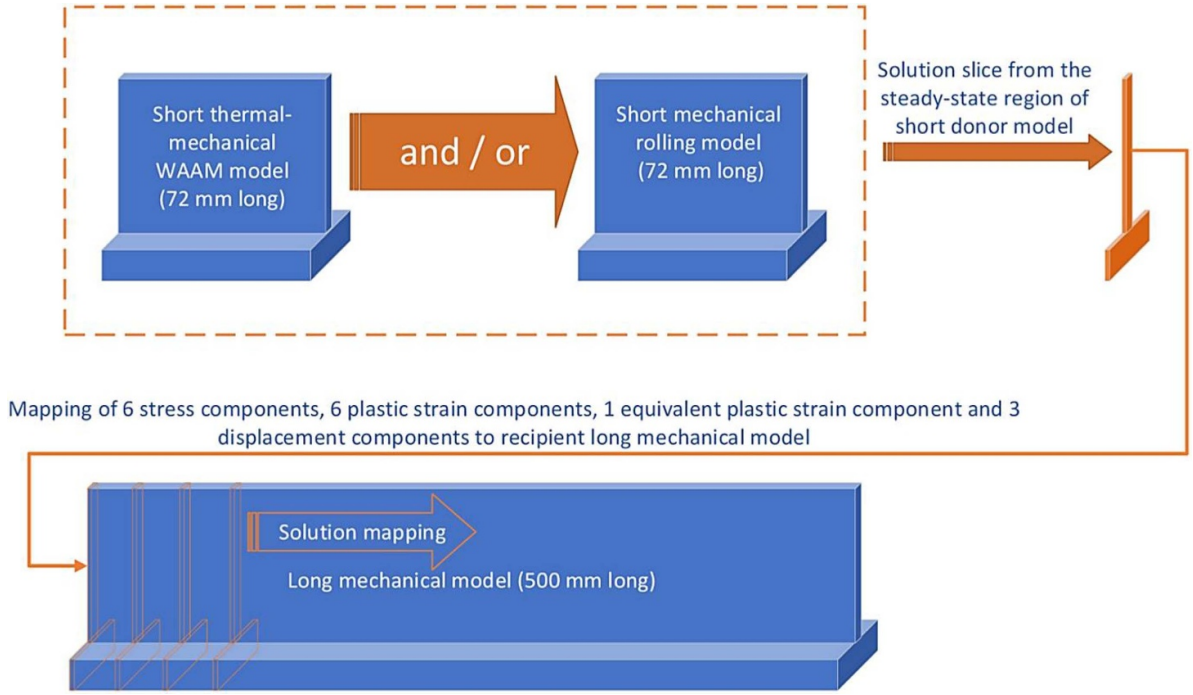


Figure 10. Schematic diagram of the hybrid approach used by Gornyakov, Sun, Ding and Williams. Reproduced from [196]. CC BY 4.0.

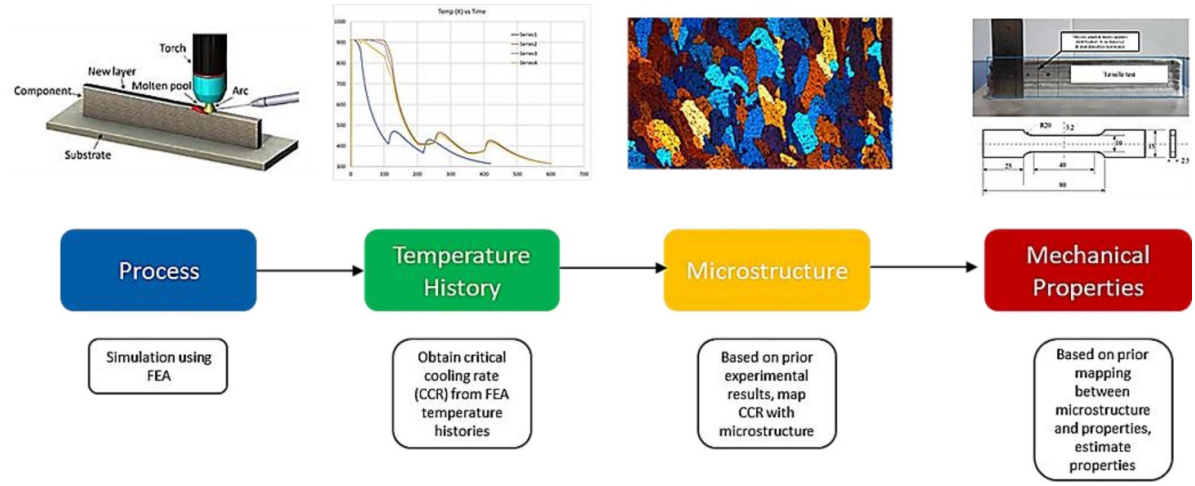


Figure 11. Illustration of the use of FEA to predict microstructures from estimated cooling rates and mechanical properties from microstructures.

and mechanical domains, and post-processing to obtain virtual temperature histories. The use of such strategies would thus hinge on assessing whether the additional effort required by the analyst may be justified. It also suffers from a lack of resolution in terms of historical data points if the modeller is interested in an in-depth analysis of the WAAM manufacturing process. Furthermore, by design, the thermal and mechanical physics were sequentially coupled, approximating the fully coupled real case. In addition, the reported works have so far investigated only simple geometries and a narrow set of process conditions, raising questions about their applicability to more complex parts. Therefore, it will be valuable to test the applicability of the hybrid methods by expanding research to include a broader set of process conditions including different materials (with dissimilar thermal properties) and part geometries.

Other long-established techniques to reduce solution times revolve around meshing strategies. These include using a graded mesh (e.g. Ding *et al* [5], where the mesh is coarser in regions where lower gradients of state variables are expected, and adaptive remeshing, (e.g. Montevecchi *et al* [197]), where the mesh is regenerated depending on the current state of the solution—again finer where larger gradients are present. Using a remeshing strategy that changed the mesh during solution, Montevecchi *et al* [197] completed the thermal–mechanical simulations in 30% of the time required to carry out

simulations without remeshing. In these instances, additional time needs to be spent upfront by the analyst to generate customized meshing strategies. Also, if the meshes need to be regenerated frequently during the adaptive remeshing solution, the additional time required could extend the solution time. Therefore, trial and error may be required to assess the optimum route, further consuming time.

In recent times, predictions of FEM models have been used to train ML models. Examples of such efforts may be found in Zhou *et al* [192], who predicted residual stresses, and Farias *et al* [24], who estimated interpass temperatures in WAAM processes.

There is a dearth of works that leverage the predicted temperature histories from FEA to correlate localized cooling rates with final microstructures at various locations. Such predictions must consider the potential for the intrinsic heat treatment that occurs due to the energy supplied when the subsequent layers are deposited in its vicinity. These microstructures may also be correlated with mechanical properties such as tensile strength and hardness, which are often direction-dependent, and bulk properties such as porosity. Studies such as this require comprehensive experimental data similar to those generated by Gierth *et al* [73] for developing robust phenomenological relationships (figure 11). Commercial FEA codes allow the use of subroutines to accomplish such tasks during the solution process.

There is a lack of experimental data for the validation of WAAM models for different combinations of materials for the part and substrate and different process conditions. A vast majority of models were developed for a given set of process parameters and were validated using experimental data generated for the same narrow set of parameters. Thus, the general applicability of the models—including the assumptions employed—could not be tested for a broad range of parameters. This provides an opportunity for experimentalists to generate extensive databases to assist in the prediction of residual stresses and distortion in WAAM under various process conditions.

Finally, even though FEA is widely used to predict residual stresses and deformations, it is worth keeping in mind that, as broached in section 4.9, it suffers from some limitations. These shortcomings include, but are not limited to, the following:

1. *The absence of modelling of fluid flow in the melt pool:* the assumption in FEA of a static melt pool without convective currents is a major approximation. For example, Hauser *et al* [202] used high-speed videography to confirm the dynamic nature of the shapes of melt pools in WAAM and that convective currents exist in the melt pool (which result in a reduction in temperature gradients within the melt pool). Furthermore, the temperature gradients themselves cause Marangoni flow within the melt pools [202], which further modifies the gradients. In FEA, no such equilibration of gradients is modelled, although some modellers (e.g. Ding *et al* [101] and Farias *et al* [24]) used an *arbitrarily* high thermal conductivity term to try to account for its effect. On the flip side, some (e.g. Wu *et al* [198]) have argued that the assumptions in FEA regarding melt pools do not introduce significant errors in the calculation of residual stresses and distortion since these effects are primarily a function of the subsequent cooling of the solid that dominates in terms of process time. Wu *et al* [198] calculated the melt pool's dimensions and its temperature profile with and without convective effects for Ti-6Al-4V and Inconel 718 and showed that there were only marginal differences (figure 12). For modelling the convective effects, they used their 'well tested heat transfer and fluid flow model' [203]. The residual stress predictions from their FEM model were successfully validated with experimental data. In contrast, the CFD study of Ou *et al* [119] demonstrated that flow in the melt pool could significantly affect the geometry of the fusion zone, as discussed in section 5.1.
Modellers will need to weigh whether the reduced solution wall time afforded by FEA over CFD and SPH would make FEA the practical choice for production or engineering situations. For more in-depth scientific studies of the melt pool and its dynamics, however, the other approaches will be required.
It may also be possible to export the temperature histories from CFD or SPH to FEM models to combine the advantages of the fluid dynamics simulations—especially in simulating the molten phase—with the power of FEM in simulating the distortion behaviour of the solid part. To the best of the authors' knowledge, no such work has been reported so far. However, one must assess if the significant additional work required in doing so (since different software codes will be employed and the geometries may not match—see the dot point immediately below) is worth the effort because the gains in accuracy may not justify it. Additionally, by design, this strategy calls for a sequentially coupled thermal–mechanical analysis. Hence the modeller must decide carefully between this route and a fully coupled thermal–mechanical FEM analysis. Such combined models are discussed further in section 6.3.
2. *The use of pre-determined shapes for additive layers:* this is arguably the largest single contributing factor for errors. The melt pool's original and changing shapes are not considered in FEA, as the pre-determined shape input into the model often corresponds to the final shape of the solidified form of the layer. This is done to ensure the new layer being added perfectly rests on the previously deposited layer to guarantee seamless contact between the elements of the two layers. In many cases, even the final forms of layers are assumed to be idealized cuboid shapes rather than the actual, often slumped, shapes. Approximations such as these introduce errors in the predictions of the model.
3. *The use of pre-determined shapes for heat source footprint and its penetration*: the analyst must assume the shape of the heat source to be used in the simulation, including the depth of its penetration. This is usually done based on experimentally obtained microstructural evidence of its effects. Additionally, thermocouple histories obtained using overhanging thermocouple installations that are built over during the WAAM process may be used; see Gierth *et al* [73]

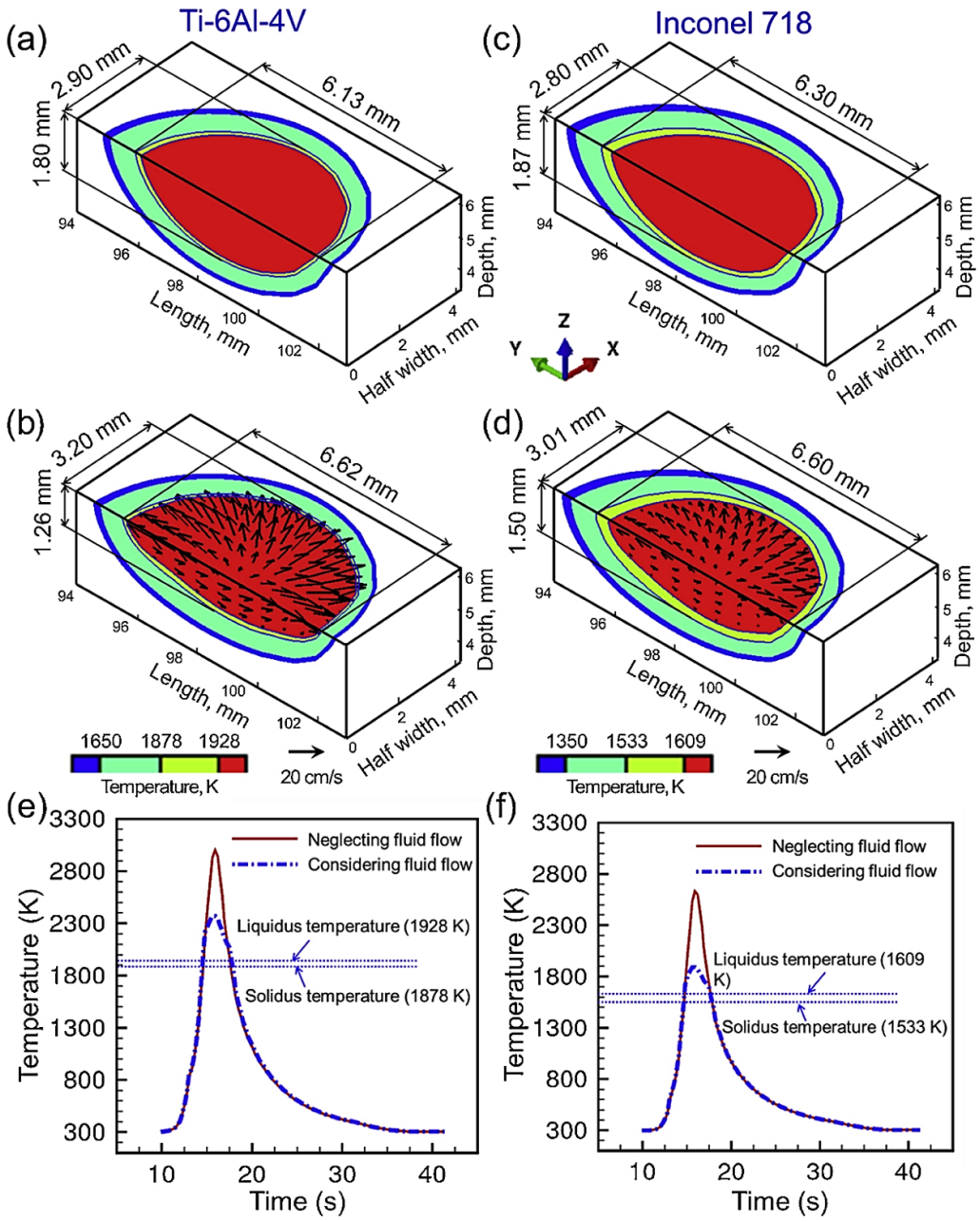


Figure 12. Temperature distribution near the molten pool for Ti-6Al-4V calculated (a) neglecting fluid flow and (b) considering fluid flow. Temperature distribution near the molten pool for Inconel 718 calculated (c) neglecting fluid flow and (d) considering fluid flow. Molten pool dimensions are indicated in the figures. Half of the molten pools are shown because of the symmetry about the *y*–*z* plane. Scanning direction is along negative *y*-axis. Comparison of temperature variation with time for 72 mm long single-track deposits of (e) Ti-6Al-4V and (f) Inconel 718 calculated considering and neglecting fluid flow. Reprinted from [198], Copyright (2021), with permission from Elsevier.

and figure 13. This task also involves several assumptions and thus introduces errors in the predictions of the model.

Future research should thus be aimed at quantifying the errors attributable to these and other limitations and, preferably, mitigating their influences on predictions. Furthermore, there is a critical need for validation data to be generated in WAAM using well-instrumented trials for temperature measurements, e.g. along the lines of Gierth *et al* [73] (as per shown in figure 13), and for part distortion and residual stress predictions.

6. Trends and opportunities in WAAM modelling

WAAM was first patented back in 1925 [4, 204], and the term 'Wire Arc Additive Manufacture' began to be used in 1984 for the purpose of scientific research [205, 206].

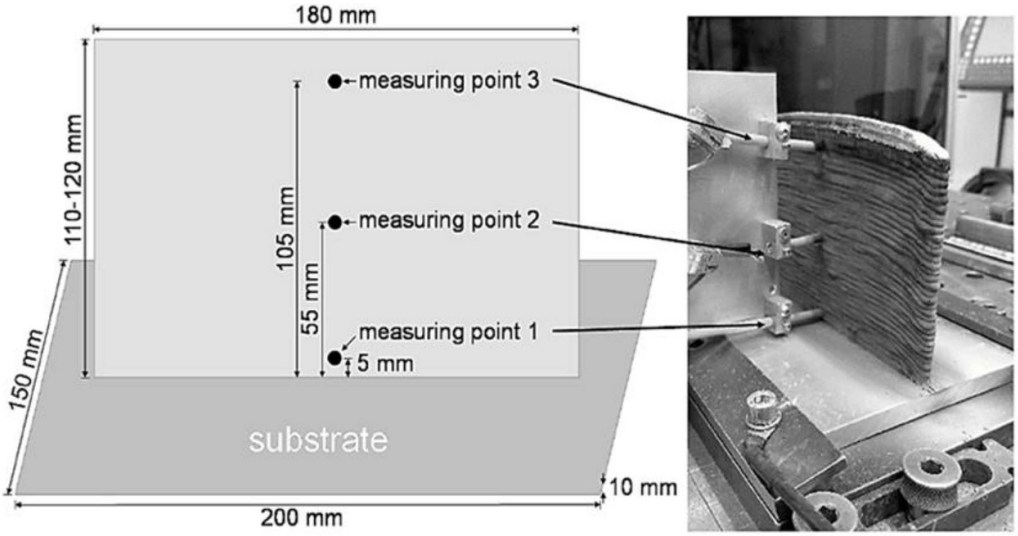


Figure 13. Instrumented experimental setup used by Gierth, Henckell, Ali, Scholl and Bergmann. Reproduced from [73]. CC BY 4.0.

WAAM has attracted interest from both industry and researchers for over 30 years. While significant progress has been made towards overcoming problems, including dimensional inaccuracy, low quality of surface finish, defect formation, and residual stress and deformation, they have not been fully resolved. Complementary approaches such as numerical modelling coupled with experimental observations to support the simulation data are proving valuable. Furthermore, cutting-edge technologies such as AI, ML and real-time sensing and monitoring are being adopted in advanced WAAM systems.

6.1. Understanding defect formation

The importance of defects and the strong research focus on their reduction was discussed in section 2.3. Combining experimental and modelling approaches allows detailed insights into the mechanisms underlying the formation of defects and guidance towards their avoidance. We consider several examples here.

Yuan *et al* [207] presented a detailed experimental investigation of the humping phenomena occurring in multi-directional robotic WAAM, demonstrating that the threshold travel speed at which humping commenced was much higher for vertically downward deposition (welding direction along the gravity direction) than horizontal (orthogonal welding direction perpendicular to the gravity direction) or vertically upward deposition (welding direction against the gravity direction). A theoretical analysis of different humping mechanisms was used to show the importance of the depression of the melt-pool surface by the arc pressure interacting with the backward metal flow under the effect of gravity. The influence of droplet impact and weld pool flow on humping in GMAW has been examined in many CFD studies (e.g. [208]), and flow simulation tools such as CFD and SPH are readily applicable to understanding humping phenomena in WAAM.

Aldalur *et al* [209] experimentally investigated the porosity and microstructure of Al–Mg alloys deposited by WAAM for three different metal transfer modes. They observed that the metal transfer mode significantly affected the porosity, with the pulsed-AC mode reducing porosity by factors of 10 and 6 compared to the standard pulsed-DC mode and the cold arc (similar to cold metal transfer) mode, respectively. Prediction of porosity using computational modelling requires specialized techniques and has mainly been applied to metal casting (e.g. [210]). However, CFD modelling can assist in the understanding of the causes of porosity. For example, Hauser *et al* [76] supplemented an experimental study of pore formation in WAAM with a CFD simulation of the shielding gas flow, allowing the influence of the flow rate on the formation and escape of gas pores to be investigated.

FEA has been used successfully to predict and ameliorate residual stress and distortion. For example, Zhao *et al* [211] used an FEA model to predict residual stress distributions and warping deformation for different deposition sequences and patterns for WAAM deposition of disc-shaped components. Srivastava *et al* [212] predicted that reducing the heat input in successive layers reduced the longitudinal residual stress by 18% compared to a constant heat input when depositing a 12-layer thin wall. In both cases, the strategies predicted by the models to reduce residual stress and distortion were verified experimentally.

In other cases, modelling has been combined with measurements to optimize process parameters rather than address specific defects. Huang *et al* [213] combined real-time x-ray observations of the flow in the melt pool with a CFD simulation of the droplet impact and melt pool flow. This allowed them to determine the process parameters that provided the preferred droplet transfer mode and optimized the melt pool flow pattern. Abe *et al* [214] used thermal sensors and a heat transfer model to determine the optimum arc voltage and wire preheating for the deposition of thin walls in

GMAW-based WAAM. The approach was demonstrated to improve the accuracy of the build.

6.2. Real-time and AI approaches

Process optimization and defect reduction can benefit from real-time monitoring and fast decision making. Accordingly, an important role of numerical modelling is to support these processes. For example, modelling can be applied prior to manufacture to determine optimum process parameters and the corresponding values of measurable parameters such as the melt pool temperature. By monitoring the parameters in real-time, as in the work of Chaurasia *et al* [215], discrepancies between the measured and predicted values can be identified and corrected following predetermined rules. For example, if the melt pool is hotter than predicted, the travel speed can be increased or the arc current reduced.

An example of this type of approach is provided by Reisch *et al* [216], who demonstrated a substantial reduction in discontinuity defects using a smart manufacturing system that detected such defects and compensated for them in subsequent layers by altering the travel speed or wire feed rate.

The use of data-driven AI algorithms is showing promise for optimizing the WAAM process and product quality. Applications investigated include defect detection [217, 218] and path planning [219]. For example, Lee *et al* [217] developed a convolutional neural network (CNN), coupled with real-time image processing, to detect defects in GTAW-based WAAM. The model detected abnormalities by capturing images of the metal transfer, melt pool and bead shapes, and the arc shape using a high dynamic range camera. The authors demonstrated that the CNN-based AI model could be the basis of a real-time monitoring tool for the WAAM process. Li *et al* [218] used a support vector ML method to identify changes in the welding voltage and current signals that point to the occurrence of defects.

Other researchers have applied AI methods for the optimization of bead shape and build geometry [220, 221] and melt-pool temperatures [222]. For example, Petrik *et al* [220] used a data-driven AI model based on a recurrent neural network [223] called AIBead to predict the shape of the bead cross-section for curved deposition paths in GMAW-based WAAM, based on the path parameters and travel speed. The model was limited to the deposition of a single layer and did not consider wire feed rates and other parameters.

A problem for data-driven AI models of WAAM is the need to gather large quantities of experimental data for training. Well-validated simulation data can potentially also be used to train such algorithms. As the examples presented at the start of this section demonstrate, coupling of experimental and modelling data is advantageous for process improvement. There is a clear opportunity for supplementing the experimental data used to train AI models with the predictions of computational models. Another prospective method is the integration of physical constraints into the AI model (physics-informed neural network—also known as PINN).

6.3. Combined and multiscale modelling approaches

WAAM is a complex process that incorporates several physical processes that occur on a wide range of scales. For example, melt pool flow and metal transfer occur on the millimetre scale, while part size can exceed 1 m. The computational grid size required to resolve the melt pool, arc and droplets are incompatible with the need to simulate the production of a large component. Moreover, the computational time required for a complete 3D simulation, including all driving phenomena, is prohibitive with current computational capabilities.

As demonstrated in section 3, different numerical modelling methods have features that provide advantages and disadvantages for different aspects of the WAAM process. Coupling of two or more methods is, consequently, widely applied in modelling as a strategy to improve the ability to capture complex multi-phase phenomena and multi-scale problems and for the purpose of formulating multi-physics efficiently (e.g. [21]). The term coupling usually implies the sharing of the computational domain between the modelling methods, with information passed from one model to another and, in some cases, in both directions (two-way coupling). Combined approaches refer to linked simulations in which the computational domain, time steps and length scales are not necessarily shared; thus, those variables are not necessarily the same for the two methods. One or more data sets calculated by one simulation are used as initial or boundary conditions for the other simulation method. Such an approach does not require running two methods simultaneously, providing more flexibility than coupled approaches while taking advantage of the strengths of the different methods and overcoming at least some of the weaknesses.

An example is the approach described by Komen *et al* [189] to treat SAW, considered in detail in section 5.2, in which CFD was used to calculate the heat flux from the arc and coupled SPH-DEM was used to model the solid and liquid domains. As discussed in sections 4.6 and 5.1–5.3, Goldak's double ellipsoidal heat flux model is widely adopted in both CFD and FEA WAAM models. Komen *et al* showed that the heat flux predicted by the CFD arc peaked at a radius of 1–2 mm from the arc axis and, therefore, differed significantly from the double ellipsoidal and other standard heat flux models.

Mu *et al* [224] pointed out that existing WAAM models fail to accurately forecast the deposition process due to difficulties in modelling the solidification process. However, SPH is well suited to the modelling of complex flows. Treatment of solidification and melting is simpler and requires fewer computational resources than mesh-based methods. Resolution can be varied in space and time, which is advantageous for WAAM modelling, in which the location of the melt pool moves continuously as a part is built. Standard SPH can realise both laminar and turbulent gas flows but is not necessarily an optimal approach for modelling gaseous and condensed phases simultaneously since the large density difference between different phases is difficult to treat. SPH has not been applied to plasmas; in principle, SPH would be well suited since it is

intrinsically compressible, but formulating boundary conditions for the solid phase would be challenging. Since SPH is not preferred for modelling the arc, the influence of the arc must be approximated by boundary conditions. Typically, analytic expressions for the heat flux and arc pressure are used, with free parameters that are determined by comparison of the predictions of the model with measurements.

On the other hand, CFD modelling of the plasma, gas and liquid phases in the arc and the melt pool has been widely adopted. A time-dependent model of the WAAM process, including the arc, wire and melt pool, is very complex. Treatment of the arc requires the inclusion of electromagnetic equations and, ideally, consideration of the production of metal vapour from the molten metal surfaces and its transport in the arc. Simulation of the formation, transfer and impact of droplets requires using the VOFs or similar methods to track the free surface between the gas and liquid phases. Practically, such a complete model is limited to, at most, a few layers and subject to an array of strong challenges. There are several possibilities for simplification. Steady-state calculations are much faster but have many limitations. Only the constant-velocity motion of the arc relative to the workpiece can be treated, so the start and end of a layer and curved deposition trajectories cannot be handled. Droplet transfer can only be treated on a time-averaged basis. Deposition of multiple layers can only be modelled if complete cooling is assumed between the deposition of each layer. A CFD model of the melt pool only, in which the arc is approximated by boundary conditions, is significantly faster and simpler but, as with SPH, requires expressions for arc and droplet heat transfer, arc pressure and, if electric current flow is included, the current density.

FEA can be used to predict the residual stress and distortion of the substrate and is well suited to large parts and multiple layers, although it does not provide flow field information. As with SPH and CFD models of the melt pool, the heat flux from the arc is generally approximated by an analytic expression.

A combined simulation approach that draws on the advantages of two or more computational methods can overcome some of the weaknesses of the individual methods. More complicated problems become feasible, and computational times can be significantly reduced.

Coupled or combined options should be based on the specific phases being modelled. For example, CFD and SPH are appropriate choices for modelling fluid problems. Since SPH is mesh-free, it is suited to treating complex geometries and free surfaces, which pose difficulties for CFD. CFD is much more widely used for modelling gas and plasma phenomena. The two simulation methods can be linked to provide a multi-phase treatment of WAAM: with CFD being used to predict the properties of the arc, which can be transferred into an SPH representation as initial or boundary conditions for the heat flux, arc pressure and (if electromagnetic calculations are included in the SPH treatment) the current density. Including complex deposition paths and multiple layers is typically easier for SPH simulations, so a combined CFD-SPH approach could be expected to provide better computational efficiency and accuracy than a single simulation method approach. Different levels of complexity would be possible.

- The simplest variant would be to run the CFD model separately for the same WAAM parameters to be used on the SPH model. An initial implementation that would provide a substantial increase in accuracy would be to use the heat flux predicted by CFD. As noted above, the heat flux can differ significantly from standard heat flux models.
- Additional accuracy will be obtained if the droplet properties and arc pressure distribution are transferred. Of these, the droplet properties (size, temperature, frequency and velocity) are the most critical since droplet impact has a substantial influence on the melt pool.
- If the SPH model includes electromagnetic effects, the current density distribution can also be transferred from the CFD arc model. The most important effect is the magnetic pinch force, which drives flow in the melt pool and can affect its depth [225]. Joule heating of the substrate is less important, especially for aluminium alloys.
- A steady-state CFD model will not be able to treat the dynamic process of droplet transfer or complex geometries but could nevertheless provide heat flux distributions from the arc that are more reliable than the analytic approximations commonly used. The steady-state model can also predict the droplet properties [225].
- A time-dependent CFD model would be the more accurate choice for predicting transient phenomena.
- The most accurate but most complicated approach would use two-way coupling between the CFD and SPH models. This would allow, for example, the influence of metal vapour and changes to the melt pool surface profile to be included dynamically in the CFD model and reflected in the heat flux to the melt pool surface transferred to the SPH model.

A CFD model of the arc can also be used to provide the heat flux data for an FEA model of the substrate, allowing a more accurate residual stress and deformation analysis. Two-way coupling is usually not of any value in this case since the FEA model does not predict metal vapour production or the shape of the melt pool surface and compute speed remains important to be able to simulate the process scale. However, if significant distortion is predicted by the FEA model, it may be worthwhile updating the substrate geometry in the CFD model since the heat transfer from the arc may be affected by the distortion.

A multi-phase, multi-physics and multi-scale model of WAAM should be possible via a combined CFD-SPH-FEA approach, in which CFD handles the arc and its interactions with the melt pool on the scale of a few millimetres, while SPH tracks the melt pool flow and solidification on a similar or slightly larger scale, and FEA is used to large scale thermal behaviour and predict residual stresses on the scale of the part.

It will be important to assess if substantial gains in accuracy may be gained using CFD and/or SPH to obtain temperature histories. While the fluid dynamics methods simulate the molten phase more accurately, it is currently not known for certain if improved estimates from these codes can significantly enhance FEM predictions.

If required, the metal microstructure can be predicted using one of several methods, including, in increasing level of

sophistication and difficulty, phenomenological, cellular automaton and phase field approaches. These methods can use the thermal history as input data and, therefore, do not have to be fully coupled, although it has been demonstrated that coupling improves the accuracy of phase field calculations in laser-powder bed fusion [226]. Phenomenological relationships between process conditions and microstructure, as well as between microstructure and properties, are relatively easy to code into FEM models to predict location-dependent material properties of WAAM parts.

Investigation of porosity formation and welding fume formation would require additional modelling techniques. While coarser porosity can be predicted using SPH, finer-scale porosity, such as from hydrogen bubble nucleation during solidification, would require, for example, the use of a discrete bubble model. Fume formation would require the inclusion of a nucleation model [55, 227], with the DEM used to track the fume particles.

6.4. Future development and application of WAAM modelling

In this section, we introduce some further examples of possible future approaches to and applications of numerical modelling that have the potential to advance WAAM research and practice.

6.4.1. Digital twins. The development of an intelligent digital twin of WAAM is a long-term goal. It requires integrating accurate predictive models of the WAAM process that run in real-time, diagnostics and monitoring of the WAAM process, and methods for active control of the process. A digital twin is continuously updated by the measurements and uses the control system to modify the process as required [83].

Mu *et al* [224] outlined the current obstacles related to building a digital twin for WAAM. They categorized these challenges into four groups: deposition modelling, scalability and efficiency, digitalization and process integration. The categories of deposition modelling and scalability and efficiency primarily encompass challenges related to numerical modelling, including the adoption of multi-scale and multi-physics approaches, development of simulation databases, understanding the influence of materials composition, and predicting solidification and the occurrence of arc blow and spatter. The combined modelling methods, as previously mentioned, are capable of effectively addressing many of these challenges. Li *et al* [228] discussed the challenges in the visualization of WAAM, which are linked to the digitalization category. They suggested that virtual reality, augmented reality (AR), and mixed reality could provide a novel way to address visualization issues in the digital twin of WAAM processes. Mu *et al* also noted the challenges in the process integration category related to the fusion of in-process sensor data, which is essential for overseeing product quality. Other problems related to sensors include calibration issues and noise reduction.

The required sensor and control technologies largely exist. Optical, electrical, acoustic and thermal sensors can be adopted, respectively, to monitor surface defects and geometrical deviations during production, detect internal defects based on spectral information, monitor the arc stability, and capture the temperature distribution along the deposited material [90]. It should be noted that interpretation of the sensor data can be difficult; for example, converting thermal radiation to temperature requires knowledge of the surface emissivity or the use of two wavelengths [229]. Control strategies that adjust process parameters, such as travel speed in real-time in response to measurements, have also already been developed. For example, Xiong and Zhang applied a vision sensor system to monitor the nozzle-to-build distance and an adaptive controller to keep this constant [230], Doumanidis and Kwak used an analytical model, coupled with measurements using infrared pyrometry and laser scanning, to control the build profile [231, 232], and Xiong *et al* demonstrated a closed-loop self-adjusting control of the layer width [233].

One of the main missing links is an accurate predictive model that is continuously updated with measured data from the physical domain as an input parameter and can run in real time. To be useful in digital twin development, the model would have to link measurable parameters, such as the melt pool temperature or dimensions, with the predicted part properties. The time horizon for the predictions would depend on the properties to be predicted; for example, dimensions of the layer under deposition would require a relatively short time horizon, while residual stress and distortion would require the complete build to be simulated. In the residual stress and distortion example, the FEA model can predict temperature distributions, but these may be inaccurate in the melt pool region since FEA does not consider molten metal flow (see section 5.2). This could be addressed by a combined CFD-FEA or SPH-FEA model, with the CFD or SPH model treating the melt pool region.

Even with advances in computational hardware and techniques, CFD, SPH and FEA models will not satisfy the real-time calculation requirement. A promising strategy is to use surrogate ML models to speed up predictions. Leiter *et al* [234] and Alowayyed *et al* [235] recently showed that a physics-agnostic surrogate model linked with single-scale models considerably reduced computational time, although the application to WAAM remains to be tested.

A problem with applying ML to complex processes such as WAAM is the difficulty and expense involved in producing large amounts of experimental data for training the model. The sparsity of data suggests that physics-informed AI models are likely to be required. Reliable computational models can address this need by providing synthetic data. For example, an FEA model can predict the residual stress and distortion for a wide range of parameters to produce a database that can be used for training an ML model. This approach, as well as being faster than manufacturing and characterizing a large number of parts, will have the further advantage of covering a larger parameter range.

6.4.2. SPH modelling of heat and mass transfer in WAAM. Even though SPH is a prominent simulation approach for heat and mass transfer problems, particularly with free surfaces and coupled other physics, it has not yet been applied to WAAM simulation. SPH is particularly suited for modelling fluid phenomena with complicated boundary dynamics, such as large boundary displacements or free surface flows. Its use in the modelling of similar processes, including arc welding and laser powder bed fusion, demonstrates the feasibility of applying SPH to WAAM. SPH is suitable for treating the deposition of multiple layers and can easily handle complex deposition paths. As discussed in section 6.3, information about the arc, including the heat flux and pressure imposed by the arc can be provided using standard analytic expressions with free parameters determined by experiment or an MHD model of the arc. An additional level of accuracy could be provided by the inclusion of electromagnetic effects in the SPH model, allowing the influence of the magnetic pinch force on melt pool flow to be considered. Finally, two-way coupling with the MHD arc model would allow the influence of metal vapour and changes to the melt pool surface profile on the arc to be considered.

6.4.3. Numerical investigation of the transient behaviour of dynamic deposition using multi-dimensional arc wires. Multi-material deposition has recently been investigated in WAAM fabrication, e.g. by Hauser *et al* [236], who produced parts with two different aluminium alloys, and Ahsan *et al* [237], who combined low carbon and stainless steels. The approach allows part properties to be targeted to the expected load distribution, for example. The possibility of a targeted material deposition from wires of different compositions and dimensions has the potential to minimize defects. Computational modelling of such a process is complicated by the need to track the mixing of the different alloys in the melt pool, considering the mixture properties, including surface tension (which affects Marangoni flow and the surface profile), viscosity, thermal conductivity, solidus and liquidus temperatures, and boiling point and heat of vaporization (which affect vaporization). The microstructure and mechanical properties of the material transition zone are also critical; for example, the formation of brittle phases and differences in the coefficients of thermal expansion can lead to cracking [76]. Despite these difficulties, transient simulation of multi-material deposition will guide the optimization of the dimensions, feed rate and feed timing of the different feed wires. CFD and SPH are both capable of simulating multi-material deposition.

6.4.4. Modelling of the production of parts with complex shapes. The primary advantage of the WAAM is its ability to manufacture parts with a wide range of geometrical features, with a substantial reduction in waste material [37]. However, experimental studies have reported that defects were observed in the L-shaped corner junction, sharp corner and square geometries, especially with Al-alloy deposition. It was identified experimentally that changes to the deposition rate and heat transfer that arose from the geometrical change significantly influenced the part quality. While FEM software is well suited to predicting the heat transfer and residual stress development during the production of complex parts, it does not consider changes to the heat transfer from the arc that occur due to the variation in substrate geometry at corners and intersections. The use of multi-physics models that allow such factors to be included will assist in the optimization of the properties of complex components.

7. Conclusions

WAAM is a metal AM process particularly suited to the rapid production of large components, attracting growing interest from industry and researchers. Nonetheless, quality problems remain, including the formation of defects and the occurrence of residual stresses and distortion. Proven numerical modelling methods such as CFD, SPH and FEM can play a significant role in addressing the problems, as they have done in the closely related process of arc welding. Modelling can also play a critical role in planning the optimum deposition strategy to ensure part quality, as part of a part qualification and process certification framework, and as a component of an intelligent digital twin.

In addition to the challenges presented by the many physical processes occurring in arc welding, WAAM modelling using physics-based approaches must deal with the large variations in spatial and temporal scales between the arc and melt pool and the part and the need to predict the deposition of many layers. While FEM can handle multi-layer deposition, its inability to realistically simulate the arc and melt-pool dynamics presents serious challenges to its reliability. Conversely, CFD and SPH can more precisely simulate the thermodynamics and dynamics of the melt pool and, in the case of CFD, also the arc, but these are limited to simulations of the deposition of a few layers because of the effect of the large-scale differences between melt pool and part scales. Combining different modelling approaches has the potential to resolve many such challenges, allowing more reliable simulation of the deposition of large parts and enabling improved prediction of their properties.

Full automation of WAAM requires linking to an intelligent digital twin, a simulated representation of the process coupled with real-time monitoring and control systems. The physics-based deterministic modelling approaches considered in this review cannot provide the real-time calculations required, so surrogate ML algorithms will be required. The physics-based models can be used to assist with training the ML models, generating synthetic data to augment experimental data, which is expensive and often difficult to obtain.

While impressive progress has been made in the computational modelling of WAAM, considerable advances are still required to meet the challenges required for modelling to assist the WAAM process to reach its full potential.

Data availability Statement

No new data were created or analysed in this study.

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